

Introduction to Electrodynamics
Fourth Edition
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All errors, typographical and substantive, and other offenses, are entirely my own.

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1 - Vector Analysis

1.1 - Vector Algebra

1.1.1 - Vector Operations

Problem: 1.1

Using the definitions in Eqs. 1.1 and 1.4, and appropriate diagrams, show that the dot product and cross product are distributive,

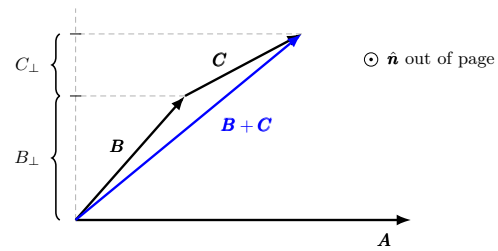
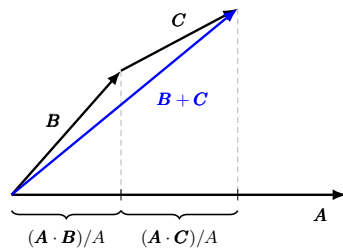
- (a) when the three vectors are coplanar;
- (b) in the general case.

The dot and cross products are defined, respectively

$$\mathbf{A} \cdot \mathbf{B} \equiv AB \cos(\theta), \quad \mathbf{A} \times \mathbf{B} \equiv AB \sin(\theta) \hat{\mathbf{n}}$$

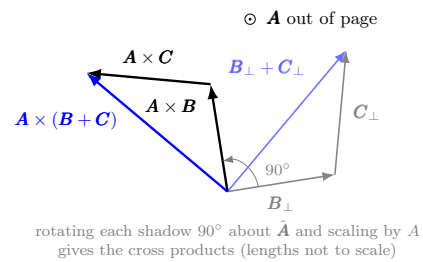
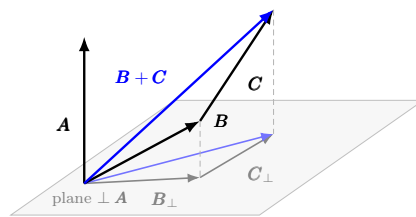
- (a) We seek to demonstrate

$$\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}, \quad \mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$$



- (b) In this part, distributivity looks like

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$$



Problem: 1.2

Is the cross product associative?

$$(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} \stackrel{?}{=} \mathbf{A} \times (\mathbf{B} \times \mathbf{C})$$

If so, *prove* it; if not, provide a counterexample (the simpler the better).

No. Consider the case where $\mathbf{A} = \mathbf{B} \neq \mathbf{0}$ and $\mathbf{C} \neq \mathbf{0}$ and $\mathbf{C} \nparallel \mathbf{A}$. Then the left-hand side is the zero vector:

$$(\mathbf{A} \times \mathbf{A}) \times \mathbf{C} = \mathbf{0}$$

but not on the right as $\mathbf{A} \times \mathbf{C} \neq \mathbf{0}$ and since $\mathbf{C} \nparallel \mathbf{A}$, we have $\mathbf{A} \nparallel (\mathbf{A} \times \mathbf{C})$. Thus

$$\mathbf{A} \times (\mathbf{A} \times \mathbf{C}) \neq \mathbf{0}$$

1.1.2 - Vector Algebra: Component Form**Problem: 1.3**

Find the angle between the body diagonals of a cube.

Let the cube have $l = w = h = a$. Situating both body diagonals at the origin, we have

$$\begin{aligned} \mathbf{d}_1 &= a\hat{x} + a\hat{y} + a\hat{z} \\ \mathbf{d}_2 &= -a\hat{x} + a\hat{y} + a\hat{z} \end{aligned}$$

Both of which have the same magnitude $\sqrt{3}a$. Computing component and projection forms, we have

$$\begin{aligned} \mathbf{d}_1 \cdot \mathbf{d}_2 &= a^2 \\ &= d_1 d_2 \cos(\theta) \\ &= 3a^2 \cos(\theta) \end{aligned}$$

implying that $\theta = \arccos(1/3) \approx 70.5^\circ$.

Problem: 1.4

Use the cross product to find the components of the unit vector $\hat{n}$ perpendicular to the shaded plane in Fig. 1.11.

One approach is to define

$$\begin{aligned} \mathbf{A} &= 0\hat{x} - 2\hat{y} + 3\hat{z} \\ \mathbf{B} &= 1\hat{x} - 2\hat{y} + 0\hat{z} \end{aligned}$$

Then the cross product is calculated as

$$\mathbf{A} \times \mathbf{B} = \det \begin{pmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & -2 & 3 \\ 1 & -2 & 0 \end{pmatrix} = 6\hat{x} + 3\hat{y} + 2\hat{z}$$

The norm of the cross product is

$$|\mathbf{A} \times \mathbf{B}| = \sqrt{6^2 + 3^2 + 2^2} = 7$$

So the vector normal to the shaded plane is

$$\hat{\mathbf{n}} = \frac{1}{7}(6\hat{\mathbf{x}} + 3\hat{\mathbf{y}} + 2\hat{\mathbf{z}})$$

1.1.3 - Triple Products

Problem: 1.5

Prove the **BAC-CAB** rule by writing out both sides in component form.

Define

$$\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$$

$$\mathbf{B} = B_x\hat{\mathbf{x}} + B_y\hat{\mathbf{y}} + B_z\hat{\mathbf{z}}$$

$$\mathbf{C} = C_x\hat{\mathbf{x}} + C_y\hat{\mathbf{y}} + C_z\hat{\mathbf{z}}$$

$$\underline{\mathbf{A} \times (\mathbf{B} \times \mathbf{C})}$$

$$\begin{aligned} \mathbf{B} \times \mathbf{C} &= \det \begin{pmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{pmatrix} \\ &= (B_y C_z - B_z C_y)\hat{\mathbf{x}} + (B_z C_x - B_x C_z)\hat{\mathbf{y}} + (B_x C_y - B_y C_x)\hat{\mathbf{z}} \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \det \begin{pmatrix} & \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & & & \\ B_y C_z - B_z C_y & B_z C_x - B_x C_z & B_x C_y - B_y C_x & \end{pmatrix} \\ &= \left[\underbrace{A_y(B_x C_y - B_y C_x)}_{x \text{ and } y} - \underbrace{A_z(B_z C_x - B_x C_z)}_{x \text{ and } z} \right] \hat{\mathbf{x}} \\ &\quad + \left[\underbrace{A_z(B_y C_z - B_z C_y)}_{y \text{ and } z} - \underbrace{A_x(B_x C_y - B_y C_x)}_{y \text{ and } x} \right] \hat{\mathbf{y}} \\ &\quad + \left[\underbrace{A_x(B_z C_x - B_x C_z)}_{z \text{ and } x} - \underbrace{A_y(B_y C_z - B_z C_y)}_{z \text{ and } y} \right] \hat{\mathbf{z}} \end{aligned}$$

The variable pairings in each vector component are bracketed so as to expose the underlying structure: for the i -th unit vector, the corresponding i -components for all of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are paired with the j - and k -components.

$$\underline{\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})}$$

$$\begin{aligned}\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) &= (A_x C_x + A_y C_y + A_z C_z)(B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}) \\ \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) &= (A_x B_x + A_y B_y + A_z B_z)(C_x \hat{\mathbf{x}} + C_y \hat{\mathbf{y}} + C_z \hat{\mathbf{z}}) \\ \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) &= \left[\underbrace{A_y(B_x C_y - B_y C_x)}_{x \text{ and } y} - \underbrace{A_z(B_z C_x - B_x C_z)}_{x \text{ and } z} \right] \hat{\mathbf{x}} \\ &\quad + \left[\underbrace{A_z(B_y C_z - B_z C_y)}_{y \text{ and } z} - \underbrace{A_x(B_x C_y - B_y C_x)}_{y \text{ and } x} \right] \hat{\mathbf{y}} \\ &\quad + \left[\underbrace{A_x(B_z C_x - B_x C_z)}_{z \text{ and } x} - \underbrace{A_y(B_y C_z - B_z C_y)}_{z \text{ and } y} \right] \hat{\mathbf{z}}\end{aligned}$$

Originally, we have eighteen terms before the subtraction. All terms with common components ($A_i B_i C_i$) vanish (three in each of $\mathbf{B}(\mathbf{A} \cdot \mathbf{C})$ and $\mathbf{C}(\mathbf{A} \cdot \mathbf{B})$). We end up with twelve terms, as we should.

Problem: 1.6

Prove that

$$[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})] + [\mathbf{B} \times (\mathbf{C} \times \mathbf{A})] + [\mathbf{C} \times (\mathbf{A} \times \mathbf{B})] = \mathbf{0}$$

Under what conditions does $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$?

Using the **BAC-CAB** identity and appealing to the commutativity of dot products, calculate

$$\begin{aligned}[\mathbf{A} \times (\mathbf{B} \times \mathbf{C})] + [\mathbf{B} \times (\mathbf{C} \times \mathbf{A})] + [\mathbf{C} \times (\mathbf{A} \times \mathbf{B})] &= \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \\ &\quad + \mathbf{C}(\mathbf{B} \cdot \mathbf{A}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C}) \\ &\quad + \mathbf{A}(\mathbf{C} \cdot \mathbf{B}) - \mathbf{B}(\mathbf{C} \cdot \mathbf{A}) \\ &= \mathbf{0}\end{aligned}$$

Equality holds when

$$\mathbf{B} \times (\mathbf{C} \times \mathbf{A}) = \mathbf{0}$$

In this case, we can write $\mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = -(\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$, establishing the special case where associativity holds.

1.1.4 - Position, Displacement, and Separation Vectors

Problem: 1.7

Find the separation vector $\mathbf{r}$ from the source point $(2, 8, 7)$ to the field point $(4, 6, 8)$. Determine its magnitude (r), and construct the unit vector $\hat{\mathbf{r}}$.

Compute

$$\mathbf{r} = (4, 6, 8) - (2, 8, 7) = (2, -2, 1)$$

which has magnitude

$$r = \sqrt{2^2 + (-2)^2 + 1^2} = \sqrt{9} = 3$$

Thus the unit vector $\hat{\mathbf{r}}$ is

$$\hat{\mathbf{r}} = \frac{2}{3}\hat{\mathbf{x}} - \frac{2}{3}\hat{\mathbf{y}} + \frac{1}{3}\hat{\mathbf{z}}$$

1.1.5 - How Vectors Transform

Problem: 1.8

- Prove that the two-dimensional rotation matrix (Eq. 1.29) preserves dot products. (That is, show that $\bar{A}_y\bar{B}_y + \bar{A}_z\bar{B}_z = A_yB_y + A_zB_z$.)
- What constraints must the elements (R_{ij}) of the three-dimensional rotation matrix (Eq. 1.30) satisfy, in order to preserve the length of $\mathbf{A}$ (for all vectors $\mathbf{A}$)?

- Using the matrix rotation

$$\begin{pmatrix} \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix}$$

We have

$$\bar{A}_y = A_y \cos(\phi) + A_z \sin(\phi)$$

$$\bar{A}_z = -A_y \sin(\phi) + A_z \cos(\phi)$$

$$\bar{B}_y = B_y \cos(\phi) + B_z \sin(\phi)$$

$$\bar{B}_z = -B_y \sin(\phi) + B_z \cos(\phi)$$

Multiply to derive

$$\begin{aligned} \bar{A}_y\bar{B}_y + \bar{A}_z\bar{B}_z &= A_yB_y \cos^2(\phi) + A_yB_z \cos(\phi) \sin(\phi) \\ &\quad + A_zB_y \cos(\phi) \sin(\phi) + A_zB_z \sin^2(\phi) \\ &\quad + A_yB_y \sin^2(\phi) - A_yB_z \cos(\phi) \sin(\phi) \\ &\quad - A_zB_y \cos(\phi) \sin(\phi) + A_zB_z \cos^2(\phi) \\ &= A_yB_y + A_zB_z \end{aligned}$$

(b) Using the matrix rotation

$$\begin{pmatrix} \bar{A}_x \\ \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{zz} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$$

We have

$$\begin{aligned} \bar{A}_x &= A_x R_{xx} + A_y R_{xy} + A_z R_{xz} \\ \bar{A}_y &= A_x R_{yx} + A_y R_{yy} + A_z R_{yz} \\ \bar{A}_z &= A_x R_{zx} + A_y R_{zy} + A_z R_{zz} \end{aligned}$$

where a transformation is length-preserving if and only if $\sum_i \bar{A}_i^2 = \sum_i A_i^2$. We ought to be succinct and write in summation form

$$\sum_i \bar{A}_i^2 = \sum_i \left(\sum_j A_j R_{ij} \right)^2$$

It is rather tedious to evaluate these quantities, but writing $\bar{A}_x^2$ out identifies the pattern:

$$\begin{aligned} \bar{A}_x^2 &= A_x R_{xx} (A_x R_{xx} + A_y R_{xy} + A_z R_{xz}) \\ &\quad + A_y R_{xy} (A_x R_{xx} + A_y R_{xy} + A_z R_{xz}) \\ &\quad + A_z R_{xz} (A_x R_{xx} + A_y R_{xy} + A_z R_{xz}) \end{aligned}$$

Equivalently, we write the summation

$$\sum_i \bar{A}_i^2 = \sum_j \sum_k \left(\sum_i R_{ij} R_{ik} \right) A_j A_k$$

Now, we need to force $\sum_j A_j^2$ on the right-hand side. One way to do so is to ensure that all products $A_j A_k$ vanish. The first condition we impose is for each $j \neq k$,

$$\sum_i R_{ij} R_{ik} = 0$$

If you are familiar with the Kronecker delta notation, we can set the sum equal to δ_{jk} . Then the last question we are behooved to ask is what of the terms *with* $j = k$. It turns out the second condition is

$$\sum_i R_{ij}^2 = 1$$

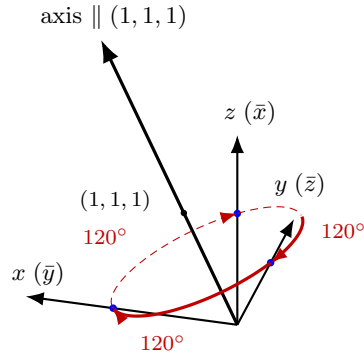
which leaves us with

$$\sum_i \bar{A}_i^2 = \sum_j A_j^2$$

Problem: 1.9

Find the transformation matrix R that describes a rotation by 120° about an axis from the origin through the point $(1, 1, 1)$. The rotation is clockwise as you look down the axis toward the origin.

An illustration is always helpful in this analysis.



rotation: 120° about the axis through $(1, 1, 1)$, clockwise as viewed looking down the axis toward O . Each unit vector slides 120° along the shared circle: $\hat{x} \rightarrow \hat{z}$, $\hat{z} \rightarrow \hat{y}$, $\hat{y} \rightarrow \hat{x}$, so the new axes are $\bar{x} \parallel z$, $\bar{y} \parallel x$, $\bar{z} \parallel y$.

The unique aspect of this situation is our axis of rotation is equidistant from each of the three axes. The axes trisect the circumference of the circular “orbit” about the axis of rotation at precisely 120° intervals – the same as the angle of rotation. Exploiting this symmetry, the key insight here is **rotation is equivalent to permutation**. The corresponding permutation matrix to the (231) rotation is

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

Problem: 1.10

- (a) How do the components of a vector transform under a **translation** of coordinates ($\bar{x} = x, \bar{y} = y - a, \bar{z} = z$, Fig. 1.16a)?
- (b) How do the components of a vector transform under an **inversion** of coordinates ($\bar{x} = -x, \bar{y} = -y, \bar{z} = -z$, Fig. 1.16b)?
- (c) How do the components of a cross product (Eq. 1.13) transform under inversion? [The cross-product of two vectors is properly called a **pseudovector** because of this “anomalous” behavior.] Is the cross product of two pseudovectors a vector, or a pseudovector? Name two pseudovector quantities in classical mechanics.
- (d) How does the scalar triple product of three vectors transform under inversions? (Such an object is called a **pseudoscalar**.)

- (a) The physical position of the vector in space from the source to field points is **unchanged**. To evaluate this, consider the invariance of displacement in the y -component:

$$\bar{A}_y = \bar{y}_2 - \bar{y}_1 = (y_2 - a) - (y_1 - a) = y_2 - y_1 = A_y$$

Therefore the coordinates of the vector are unchanged.

- (b) Once more, the physical position of the vector in space is invariant, but to reflect the inversion of the axes in preserving position, all components undergo a sign change:

$$\bar{A}_i = -A_i$$

- (c) The anomaly lies in the fact that the components of a cross product **do not** change under inversion, even if the sign of the components to the inputs **A** and **B** vectors *did* flip. Put differently, **A** and **B** **do not** actually change their position in space, as it is only the axes that undergo inversion, and the signs of the components flip to reflect that. Concretely,

$$\begin{aligned}
 \bar{\mathbf{A}} &= \bar{A}_x \hat{\mathbf{x}} + \bar{A}_y \hat{\mathbf{y}} + \bar{A}_z \hat{\mathbf{z}} \\
 &= -A_x \hat{\mathbf{x}} - A_y \hat{\mathbf{y}} - A_z \hat{\mathbf{z}} \\
 \bar{\mathbf{B}} &= \bar{B}_x \hat{\mathbf{x}} + \bar{B}_y \hat{\mathbf{y}} + \bar{B}_z \hat{\mathbf{z}} \\
 &= -B_x \hat{\mathbf{x}} - B_y \hat{\mathbf{y}} - B_z \hat{\mathbf{z}} \\
 \bar{\mathbf{A}} \times \bar{\mathbf{B}} &= \underbrace{[(-A_y)(-B_z) - (-A_z)(-B_y)]}_{\text{sign cancellation}} \hat{\mathbf{x}} + (A_z B_x - A_x B_z) \hat{\mathbf{y}} + (A_x B_y - A_y B_x) \hat{\mathbf{z}} \\
 &= \mathbf{A} \times \mathbf{B}
 \end{aligned}$$

Therein that derivation, we find a concise definition for a pseudovector: under inversion, we have

$$\bar{\mathbf{A}} = \mathbf{A}$$

Similarly, we consider two pseudovectors **C** and **D**. Inverting to $\bar{\mathbf{C}}$ and $\bar{\mathbf{D}}$, by the same logic as above, the cross product of two pseudovectors is itself

another pseudovector. Lastly, angular velocity and angular momentum are two pseudovector quantities in classical mechanics. Recall that the velocity of a point in a rotating body is given by

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$$

where $\boldsymbol{\omega}$ is the angular velocity and $\mathbf{r}$ the radial vector. Now invert all of the vectors to derive:

$$-\mathbf{v} = \bar{\boldsymbol{\omega}} \times (-\mathbf{r}) = -(\bar{\boldsymbol{\omega}} \times \mathbf{r}) \implies \mathbf{v} = \bar{\boldsymbol{\omega}} \times \mathbf{r}$$

We have demonstrated that $\bar{\boldsymbol{\omega}} = \boldsymbol{\omega}$ under inversion, ascertaining its pseudovector quality. For angular momentum $\mathbf{L}$, recall that the definition

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

lends itself to our previous derivation of $\bar{\mathbf{A}} \times \bar{\mathbf{B}} = \mathbf{A} \times \mathbf{B}$; simply replace the variables to see angular momentum is too a pseudovector.

- (d) Under inversion, $\bar{\mathbf{A}} = -\mathbf{A}$ for any (real) vector $\mathbf{A}$. Derive

$$\begin{aligned} \bar{\mathbf{A}} \cdot (\bar{\mathbf{B}} \times \bar{\mathbf{C}}) &= \bar{\mathbf{A}} \cdot (\mathbf{B} \times \mathbf{C}) \\ &= -\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) \end{aligned}$$

Unlike in the case of the pseudovector, the pseudoscalar **does** change sign.

1.2 - Differential Calculus

1.2.2 - Gradient

Problem: 1.11

Find the gradients of the following functions:

(a) $f(x, y, z) = x^2 + y^3 + z^4$

(b) $f(x, y, z) = x^2 y^3 z^4$

(c) $f(x, y, z) = e^x \sin(y) \ln(z)$

(a) $\nabla f = 2x\hat{\mathbf{x}} + 3y^2\hat{\mathbf{y}} + 4z^3\hat{\mathbf{z}}$

(b) $\nabla f = 2xy^3z^4\hat{\mathbf{x}} + 3x^2y^2z^4\hat{\mathbf{y}} + 4x^2y^3z^3\hat{\mathbf{z}}$

(c) $\nabla f = e^x [\sin(y) \ln(z) \hat{\mathbf{x}} + \cos(y) \ln(z) \hat{\mathbf{y}} + \sin(y) \frac{1}{z} \hat{\mathbf{z}}]$

Problem: 1.12

The height of a certain hill (in feet) is given by

$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$$

where y is the distance (in miles) north, x the distance east of South Hadley.

(a) Where is the top of the hill located?

(b) How high is the hill?

(c) How steep is the slope (in feet per mile) at a point 1 mile north and one mile east of South Hadley? In what direction is the slope steepest, at that point?

(a) Calculate

$$\nabla h(x, y) = 10[(2y - 6x - 18)\hat{\mathbf{x}} + (2x - 8y + 28)\hat{\mathbf{y}}]$$

An extremum of the hill is located at (x, y) satisfying

$$\begin{pmatrix} -6 & 2 \\ 2 & -8 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 18 \\ -28 \end{pmatrix}$$

Which is solved by $(-2, 3)$. The Hessian and its determinant are given by

$$D = 10^2 \cdot \det \begin{pmatrix} -6 & 2 \\ 2 & -8 \end{pmatrix} = 4400$$

Since $D > 0$ and $f_{xx} = -6 < 0$, we can confirm that this is a local maximum.

(b)

$$h(-2, 3) = 10 \left[2(-2)(3) - 3(-2)^2 - 4(3)^2 - 18(-2) + 28(3) + 12 \right] = 72 \text{ feet}$$

(c) Let $x = 1$ and $y = 1$. Then

$$\begin{aligned} |\nabla h(1, 1)| &= \sqrt{10(2(1) - 6(1) - 18)^2 + 10(2(1) - 8(1) + 28)^2} \\ &= \sqrt{(-220)^2 + 220^2} = 220\sqrt{2} \\ &\approx 311 \text{ ft/mi} \end{aligned}$$

and the direction of the steepest slope points towards

$$\nabla h(1, 1) = -220\hat{\mathbf{x}} + 220\hat{\mathbf{y}}$$

Problem: 1.13

Let $\mathbf{r}$ be the separation vector from a fixed point (x', y', z') to the point (x, y, z) and let r be its length. Show that

- (a) $\nabla(r^2) = 2\mathbf{r}$
- (b) $\nabla(1/r) = -\hat{\mathbf{r}}/r^2$
- (c) What is the *general* formula for $\nabla(r^n)$?

For each problem, make use of the following results:

$$\frac{\partial r}{\partial i} = \frac{i - i'}{r}, \quad i \in \{x, y, z\}$$

(a) The squared length of $\mathbf{r}$ is

$$r^2 = (x - x')^2 + (y - y')^2 + (z - z')^2$$

Then its gradient is

$$\begin{aligned} \nabla(r^2) &= 2r \left(\frac{\partial r}{\partial x} \hat{\mathbf{x}} + \frac{\partial r}{\partial y} \hat{\mathbf{y}} + \frac{\partial r}{\partial z} \hat{\mathbf{z}} \right) \\ &= 2r \left[\frac{(x - x')}{r} \hat{\mathbf{x}} + \frac{(y - y')}{r} \hat{\mathbf{y}} + \frac{(z - z')}{r} \hat{\mathbf{z}} \right] \\ &= 2[(x - x') \hat{\mathbf{x}} + (y - y') \hat{\mathbf{y}} + (z - z') \hat{\mathbf{z}}] \\ &= 2\mathbf{r} \end{aligned}$$

(b) The inverse length is

$$\frac{1}{r} = \left[(x - x')^2 + (y - y')^2 + (z - z')^2 \right]^{-1/2}$$

From this we calculate

$$\begin{aligned}\nabla\left(\frac{1}{r}\right) &= -\frac{1}{r^2}\left(\frac{\partial r}{\partial x}\hat{\mathbf{x}} + \frac{\partial r}{\partial y}\hat{\mathbf{y}} + \frac{\partial r}{\partial z}\hat{\mathbf{z}}\right) \\ &= -\frac{1}{r^2}\left[\frac{(x-x')}{r}\hat{\mathbf{x}} + \frac{(y-y')}{r}\hat{\mathbf{y}} + \frac{(z-z')}{r}\hat{\mathbf{z}}\right] \\ &= -\frac{\hat{\mathbf{r}}}{r^2}\end{aligned}$$

(c) For $n > 0$, derive each component

$$\frac{\partial}{\partial i}(r^n) = n r^{n-1} \frac{\partial r}{\partial i}$$

which produces gradient

$$\nabla(r^n) = n r^{n-1} \hat{\mathbf{r}}$$

Analogously, in the inverse case, each component is derived as

$$\frac{\partial}{\partial i}\left(\frac{1}{r^n}\right) = -\frac{n}{r^{n+1}} \frac{\partial r}{\partial i}$$

which corresponds to gradient

$$\nabla\left(\frac{1}{r^n}\right) = -\frac{n}{r^{n+1}} \hat{\mathbf{r}}$$

We may actually combine these formulae for all $n \in \mathbb{R}$, and simply write

$$\nabla(r^n) = n r^{n-1} \hat{\mathbf{r}}$$

Problem: 1.14

Suppose that f is a function of two variables (y and z) only. Show that the gradient $\nabla f = (\partial f/\partial y)\hat{\mathbf{y}} + (\partial f/\partial z)\hat{\mathbf{z}}$ transforms as a vector under rotations, Eq. 1.29. [Hint: $(\partial f/\partial \bar{y}) = (\partial f/\partial y)(\partial y/\partial \bar{y}) + (\partial f/\partial z)(\partial z/\partial \bar{y})$, and the analogous formula for $\partial f/\partial \bar{z}$. We know that $\bar{y} = y \cos(\phi) + z \sin(\phi)$ and $\bar{z} = -y \sin(\phi) + z \cos(\phi)$; “solve” these equations for y and z (as functions of $\bar{y}$ and $\bar{z}$, and compute the needed derivatives $\partial y/\partial \bar{y}$, $\partial z/\partial \bar{y}$, etc.)]

The transformed coordinates $\bar{y}$ and $\bar{z}$ are defined as

$$\begin{pmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{pmatrix} \begin{pmatrix} y \\ z \end{pmatrix} = \begin{pmatrix} \bar{y} \\ \bar{z} \end{pmatrix}$$

and y and z are solved for by

$$\begin{aligned}\begin{pmatrix} y \\ z \end{pmatrix} &= \begin{pmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{pmatrix} \begin{pmatrix} \bar{y} \\ \bar{z} \end{pmatrix} \\ y &= \bar{y} \cos(\phi) - \bar{z} \sin(\phi) \\ z &= \bar{y} \sin(\phi) + \bar{z} \cos(\phi)\end{aligned}$$

The gradient in the transformed coordinate space takes on form

$$\nabla f = \frac{\partial f}{\partial \bar{y}} \hat{\mathbf{y}} + \frac{\partial f}{\partial \bar{z}} \hat{\mathbf{z}}$$

where by the chain rule

$$\begin{aligned} \frac{\partial f}{\partial \bar{y}} &= \frac{\partial f}{\partial y} \frac{\partial y}{\partial \bar{y}} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial \bar{y}} \\ \frac{\partial f}{\partial \bar{z}} &= \frac{\partial f}{\partial y} \frac{\partial y}{\partial \bar{z}} + \frac{\partial f}{\partial z} \frac{\partial z}{\partial \bar{z}} \end{aligned}$$

From our equations for y and z , we calculate the partial derivatives with respect to the transformed coordinates as

$$\frac{\partial y}{\partial \bar{y}} = \cos(\phi), \quad \frac{\partial y}{\partial \bar{z}} = -\sin(\phi), \quad \frac{\partial z}{\partial \bar{y}} = \sin(\phi), \quad \frac{\partial z}{\partial \bar{z}} = \cos(\phi)$$

Combining our results into the rotation matrix and corresponding matrix equation, we conclude that the gradient transforms as a vector under rotations; particularly, the rotation matrix is precisely the same as that in the transform from (y, z) to $(\bar{y}, \bar{z})$.

$$\begin{pmatrix} \partial f / \partial \bar{y} \\ \partial f / \partial \bar{z} \end{pmatrix} = \begin{pmatrix} \cos(\phi) & \sin(\phi) \\ -\sin(\phi) & \cos(\phi) \end{pmatrix} \begin{pmatrix} \partial f / \partial y \\ \partial f / \partial z \end{pmatrix}$$

1.2.4 - The Divergence

Problem: 1.15

Calculate the divergence of the following vector functions:

- (a) $\mathbf{v}_a = x^2 \hat{\mathbf{x}} + 3xz^2 \hat{\mathbf{y}} - 2xz \hat{\mathbf{z}}$
- (b) $\mathbf{v}_b = xy \hat{\mathbf{x}} + 2yz \hat{\mathbf{y}} + 3zx \hat{\mathbf{z}}$
- (c) $\mathbf{v}_c = y^2 \hat{\mathbf{x}} + (2xy + z^2) \hat{\mathbf{y}} + 2yz \hat{\mathbf{z}}$

$$(a) \quad \nabla \cdot \mathbf{v}_a = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(3xz^2) - \frac{\partial}{\partial z}(2xz) = 2x - 2x = 0$$

$$(b) \quad \nabla \cdot \mathbf{v}_b = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(2yz) + \frac{\partial}{\partial z}(3zx) = 3x + y + 2z$$

$$(c) \quad \mathbf{v}_c = \frac{\partial}{\partial x}(y^2) + \frac{\partial}{\partial y}(2xy + z^2) + \frac{\partial}{\partial z}(2yz) = 2x + 2y$$

Problem: 1.16

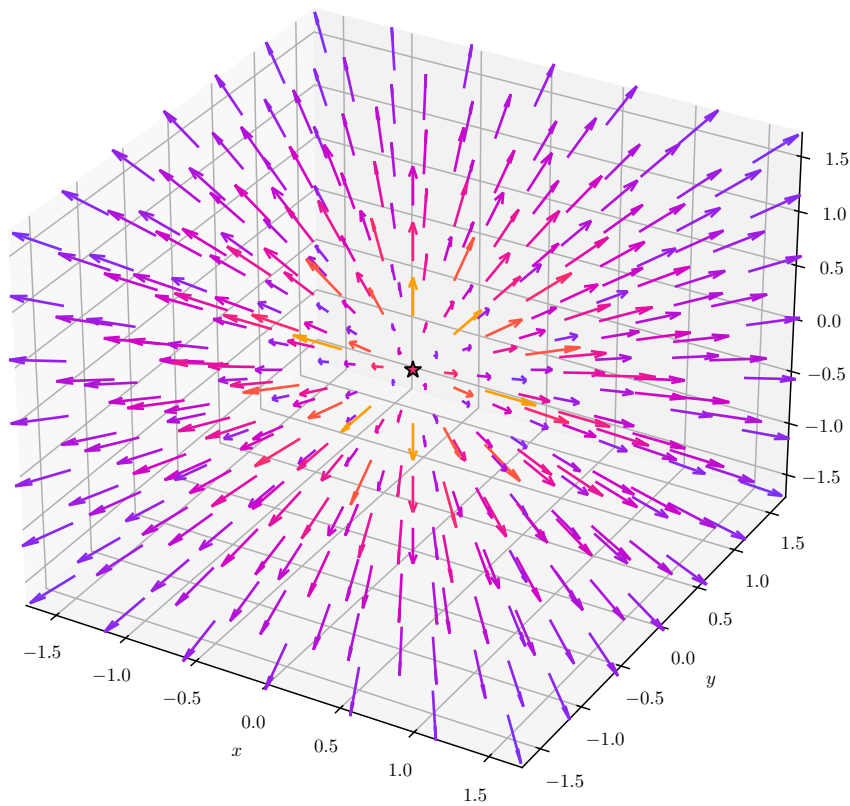
Sketch the vector function

$$\mathbf{v} = \frac{\hat{\mathbf{r}}}{r^2}$$

and compute its divergence. The answer may surprise you...can you explain it?

We graph the vector function below:

$$\mathbf{v} = \hat{\mathbf{r}}/r^2 \text{ in 3D (color} = \log_{10} |\mathbf{v}|)$$



Let

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \hat{\mathbf{r}} = \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{\sqrt{x^2 + y^2 + z^2}}$$

Then

$$\mathbf{v} = \frac{\hat{\mathbf{r}}}{r^2} = \frac{x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}}{(x^2 + y^2 + z^2)^{3/2}}$$

The divergence is computed as

$$\begin{aligned}
 \nabla \cdot \mathbf{v} &= \frac{\partial}{\partial x} \left(\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \right) + \frac{\partial}{\partial y} \left(\frac{y}{(x^2 + y^2 + z^2)^{3/2}} \right) \\
 &\quad + \frac{\partial}{\partial z} \left(\frac{z}{(x^2 + y^2 + z^2)^{3/2}} \right) \\
 &= \frac{(y^2 + z^2 - 2x^2) + (x^2 + z^2 - 2y^2) + (x^2 + y^2 - 2z^2)}{(x^2 + y^2 + z^2)^{5/2}} \\
 &= 0
 \end{aligned}$$

The counterintuitive solution, then, is the radial vector $\mathbf{v}$ has **vanishing divergence everywhere except the origin**, for this would produce $0/0$. The mechanics of the delta function later will enable us to explore this quandary deeper.

Problem: 1.17

In two dimensions, show that the divergence transforms as a scalar under rotations. [*Hint*: Use Eq. 1.29 to determine $\bar{v}_y$ and $\bar{v}_z$, and the method of Prob. 1.14 to calculate the derivatives. Your aim is to show that $\partial \bar{v}_y / \partial \bar{y} + \partial \bar{v}_z / \partial \bar{z} = \partial v_y / \partial y + \partial v_z / \partial z$.]

By our rotation matrix equation, the transformed quantities are

$$\begin{aligned}
 \bar{v}_y &= v_y \cos(\phi) + v_z \sin(\phi) \\
 \bar{v}_z &= -v_y \sin(\phi) + v_z \cos(\phi)
 \end{aligned}$$

The inverse rotation takes us back to

$$\begin{aligned}
 v_y &= \bar{v}_y \cos(\phi) - \bar{v}_z \sin(\phi) \\
 v_z &= \bar{v}_y \sin(\phi) + \bar{v}_z \cos(\phi)
 \end{aligned}$$

Using the chain rule, the derivation for $\partial \bar{v}_y / \partial \bar{y}$ and $\partial \bar{v}_z / \partial \bar{z}$ is

$$\begin{aligned}
 \frac{\partial \bar{v}_y}{\partial \bar{y}} &= \frac{\partial \bar{v}_y}{\partial v_y} \left(\frac{\partial v_y}{\partial y} \frac{\partial y}{\partial \bar{y}} + \frac{\partial v_y}{\partial z} \frac{\partial z}{\partial \bar{y}} \right) + \frac{\partial \bar{v}_y}{\partial v_z} \left(\frac{\partial v_z}{\partial y} \frac{\partial y}{\partial \bar{y}} + \frac{\partial v_z}{\partial z} \frac{\partial z}{\partial \bar{y}} \right) \\
 \frac{\partial \bar{v}_z}{\partial \bar{z}} &= \frac{\partial \bar{v}_z}{\partial v_y} \left(\frac{\partial v_y}{\partial y} \frac{\partial y}{\partial \bar{z}} + \frac{\partial v_y}{\partial z} \frac{\partial z}{\partial \bar{z}} \right) + \frac{\partial \bar{v}_z}{\partial v_z} \left(\frac{\partial v_z}{\partial y} \frac{\partial y}{\partial \bar{z}} + \frac{\partial v_z}{\partial z} \frac{\partial z}{\partial \bar{z}} \right)
 \end{aligned}$$

From problem 1.14, we have each of $\partial y / \partial \bar{y}$, $\partial y / \partial \bar{z}$, $\partial z / \partial \bar{y}$, and $\partial z / \partial \bar{z}$. From the rotation equations above, we also have

$$\frac{\partial \bar{v}_y}{\partial v_y} = \cos(\phi), \quad \frac{\partial \bar{v}_y}{\partial v_z} = \sin(\phi), \quad \frac{\partial \bar{v}_z}{\partial v_y} = -\sin(\phi), \quad \frac{\partial \bar{v}_z}{\partial v_z} = \cos(\phi)$$

Combining our ingredients and invoking the Pythagorean theorem, conclude

$$\begin{aligned}
 \frac{\partial \bar{v}_y}{\partial \bar{y}} &= \frac{\partial v_y}{\partial y} \cos^2(\phi) + \frac{\partial v_y}{\partial z} \sin(\phi) \cos(\phi) \\
 &\quad + \frac{\partial v_z}{\partial y} \sin(\phi) \cos(\phi) + \frac{\partial v_z}{\partial z} \sin^2(\phi) \\
 \frac{\partial \bar{v}_z}{\partial \bar{z}} &= \frac{\partial v_z}{\partial z} \cos^2(\phi) - \frac{\partial v_y}{\partial z} \sin(\phi) \cos(\phi) \\
 &\quad - \frac{\partial v_z}{\partial y} \sin(\phi) \cos(\phi) + \frac{\partial v_y}{\partial y} \sin^2(\phi) \\
 \implies \frac{\partial \bar{v}_y}{\partial \bar{y}} + \frac{\partial \bar{v}_z}{\partial \bar{z}} &= \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}
 \end{aligned}$$

1.2.5 - The Curl

Problem: 1.18

Calculate the curls of the vector functions in Prob. 1.15.

(a)

$$\begin{aligned}
 \nabla \times \mathbf{v}_a &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & 3xz^2 & -2xz \end{vmatrix} \\
 &= \left[\frac{\partial}{\partial y}(-2xz) - \frac{\partial}{\partial z}(3xz^2) \right] \hat{\mathbf{x}} + \left[\frac{\partial}{\partial z}(x^2) - \frac{\partial}{\partial x}(-2xz) \right] \hat{\mathbf{y}} \\
 &\quad + \left[\frac{\partial}{\partial x}(3xz^2) - \frac{\partial}{\partial y}(x^2) \right] \hat{\mathbf{z}} \\
 &= -6xz\hat{\mathbf{x}} + 2z\hat{\mathbf{y}} + 3z^2\hat{\mathbf{z}}
 \end{aligned}$$

(b)

$$\begin{aligned}
 \nabla \times \mathbf{v}_b &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & 2yz & 3zx \end{vmatrix} \\
 &= \left[\frac{\partial}{\partial y}(3zx) - \frac{\partial}{\partial z}(2yz) \right] \hat{\mathbf{x}} + \left[\frac{\partial}{\partial z}(xy) - \frac{\partial}{\partial x}(3zx) \right] \hat{\mathbf{y}} \\
 &\quad + \left[\frac{\partial}{\partial x}(2yz) - \frac{\partial}{\partial y}(xy) \right] \hat{\mathbf{z}} \\
 &= -2y\hat{\mathbf{x}} - 3z\hat{\mathbf{y}} - x\hat{\mathbf{z}}
 \end{aligned}$$

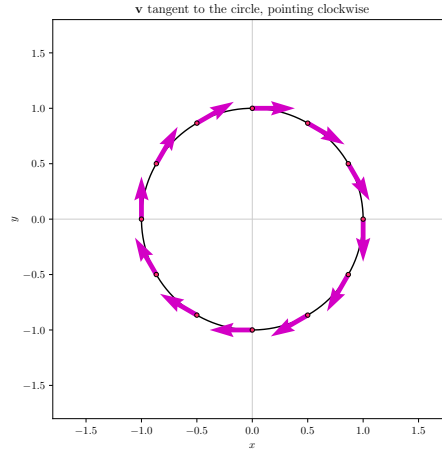
(c)

$$\begin{aligned}
\nabla \times \mathbf{v}_c &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & 2xy + z^2 & 2yz \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y}(2yz) - \frac{\partial}{\partial z}(2xy + z^2) \right] \hat{\mathbf{x}} + \left[\frac{\partial}{\partial z}(y^2) - \frac{\partial}{\partial x}(2yz) \right] \hat{\mathbf{y}} \\
&\quad + \left[\frac{\partial}{\partial x}(2xy + z^2) - \frac{\partial}{\partial y}(y^2) \right] \hat{\mathbf{z}} \\
&= \mathbf{0}
\end{aligned}$$

Problem: 1.19

Draw a circle in the xy plane. At a few representative points draw the vector $\mathbf{v}$ tangent to the circle, pointing in the clockwise direction. By comparing adjacent vectors, determine the *sign* of $\partial v_x/\partial y$ and $\partial v_y/\partial x$. According to Eq. 1.41, then, what is the direction of $\nabla \times \mathbf{v}$? Explain how this example illustrates the geometrical interpretation of the curl.

We illustrate tangent vectors pointing clockwise on the unit circle:



Geometrically, observe how as x increases, the v_y component decreases, and as y increases, v_x increases. Thus $\partial v_x/\partial y > 0$ and $\partial v_y/\partial x < 0$.

Analytically, the vector field describing these tangent clockwise vectors is

$$\mathbf{v} = v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} = y\hat{\mathbf{x}} - x\hat{\mathbf{y}}$$

Meaning that $\partial v_x/\partial y > 0$ and $\partial v_y/\partial x < 0$, and thus $\partial v_y/\partial x - \partial v_x/\partial y < 0$. This suggests that the curl points in the **negative** z -direction.

$$\nabla \times \mathbf{v} = \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}} = -2\hat{\mathbf{z}}$$

By the right-handedness of the coordinate system, this is geometrically intuitive. From our computation of $\nabla \times \mathbf{v}$, we see that the curl is in the negative z -direction

everywhere, indicating that if we took any point and applied the right-hand rule, our thumb would point downwards everywhere.

Problem: 1.20

Construct a vector function that has zero divergence and zero curl everywhere. (A *constant* will do the job, of course, but make it something a little more interesting than that!)

Let $\mathbf{v} = v_x \hat{\mathbf{x}} + v_y \hat{\mathbf{y}} + v_z \hat{\mathbf{z}}$. We must find a vector $\mathbf{v}$ whose components satisfy these relations:

$$\begin{aligned}\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} &= 0 \\ \frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} &= 0 \\ \frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} &= 0 \\ \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} &= 0\end{aligned}$$

One insight is to realize that if each of the three components are purely in terms of x , y , and z , respectively, then immediately the curl goes to zero. Suppose $v_x = f(x)$, $v_y = g(y)$, and $v_z = h(z)$. Then we need only find functions that satisfy

$$\frac{df}{dx} + \frac{dg}{dy} + \frac{dh}{dz} = 0$$

An easy example is: $f(x) = -x$, $g(y) = -y$, and $h(z) = 2z$.

1.2.6 - Product Rules

Problem: 1.21

Prove product rules (i), (iv), and (v).

$$(i) \nabla(fg) = f\nabla g + g\nabla f.$$

PROOF.

$$\begin{aligned}\nabla(fg) &= \frac{\partial}{\partial x}(fg) \hat{\mathbf{x}} + \frac{\partial}{\partial y}(fg) \hat{\mathbf{y}} + \frac{\partial}{\partial z}(fg) \hat{\mathbf{z}} \\ &= \left(f \frac{\partial g}{\partial x} + g \frac{\partial f}{\partial x}\right) \hat{\mathbf{x}} + \left(f \frac{\partial g}{\partial y} + g \frac{\partial f}{\partial y}\right) \hat{\mathbf{y}} + \left(f \frac{\partial g}{\partial z} + g \frac{\partial f}{\partial z}\right) \hat{\mathbf{z}} \\ &= f\nabla g + g\nabla f\end{aligned}$$

□

$$(iv) \nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}).$$

PROOF. First compute $\mathbf{A} \times \mathbf{B}$:

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \\ &= (A_y B_z - A_z B_y) \hat{\mathbf{x}} + (A_z B_x - A_x B_z) \hat{\mathbf{y}} + (A_x B_y - A_y B_x) \hat{\mathbf{z}}\end{aligned}$$

Then

$$\begin{aligned}\nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \frac{\partial}{\partial x}(A_y B_z - A_z B_y) + \frac{\partial}{\partial y}(A_z B_x - A_x B_z) + \frac{\partial}{\partial z}(A_x B_y - A_y B_x) \\ &= A_y \frac{\partial B_z}{\partial x} + B_z \frac{\partial A_y}{\partial x} - A_z \frac{\partial B_y}{\partial x} - B_y \frac{\partial A_z}{\partial x} \\ &\quad + A_z \frac{\partial B_x}{\partial y} + B_x \frac{\partial A_z}{\partial y} - A_x \frac{\partial B_z}{\partial y} - B_z \frac{\partial A_x}{\partial y} \\ &\quad + A_x \frac{\partial B_y}{\partial z} + B_y \frac{\partial A_x}{\partial z} - A_y \frac{\partial B_x}{\partial z} - B_x \frac{\partial A_y}{\partial z} \\ &= \underbrace{B_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + B_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + B_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)}_{\mathbf{B}(\nabla \times \mathbf{A})} \\ &\quad + \underbrace{A_x \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) + A_y \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) + A_z \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right)}_{\mathbf{A}(\nabla \times \mathbf{B})}\end{aligned}$$

□

$$(v) \nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f).$$

PROOF.

$$\begin{aligned}\nabla \times (f\mathbf{A}) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ fA_x & fA_y & fA_z \end{vmatrix} \\ &= \left[\frac{\partial}{\partial y}(fA_z) - \frac{\partial}{\partial z}(fA_y) \right] \hat{\mathbf{x}} + \left[\frac{\partial}{\partial z}(fA_x) - \frac{\partial}{\partial x}(fA_z) \right] \hat{\mathbf{y}} \\ &\quad + \left[\frac{\partial}{\partial x}(fA_y) - \frac{\partial}{\partial y}(fA_x) \right] \hat{\mathbf{z}} \\ &= \left[f \frac{\partial A_z}{\partial y} + A_z \frac{\partial f}{\partial y} - f \frac{\partial A_y}{\partial z} - A_y \frac{\partial f}{\partial z} \right] \hat{\mathbf{x}} \\ &\quad + \left[f \frac{\partial A_x}{\partial z} + A_x \frac{\partial f}{\partial z} - f \frac{\partial A_z}{\partial x} - A_z \frac{\partial f}{\partial x} \right] \hat{\mathbf{y}} \\ &\quad + \left[f \frac{\partial A_y}{\partial x} + A_y \frac{\partial f}{\partial x} - f \frac{\partial A_x}{\partial y} - A_x \frac{\partial f}{\partial y} \right] \hat{\mathbf{z}} \\ &= f \left[\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{\mathbf{z}} \right] \\ &\quad - \underbrace{\left[\left(A_y \frac{\partial f}{\partial z} - A_z \frac{\partial f}{\partial y} \right) \hat{\mathbf{x}} + \left(A_x \frac{\partial f}{\partial z} - A_z \frac{\partial f}{\partial x} \right) \hat{\mathbf{y}} + \left(A_x \frac{\partial f}{\partial y} - A_y \frac{\partial f}{\partial x} \right) \hat{\mathbf{z}} \right]}_{\mathbf{A} \times (\nabla f)}\end{aligned}$$

□

Problem: 1.22

- (a) If $\mathbf{A}$ and $\mathbf{B}$ are two vector functions, what does the expression $(\mathbf{A} \cdot \nabla) \mathbf{B}$ mean? (That is, what are its x , y , and z components, in terms of the Cartesian components of $\mathbf{A}$, $\mathbf{B}$, and ∇ ?)
- (b) Compute $(\hat{\mathbf{r}} \cdot \nabla) \hat{\mathbf{r}}$, where $\hat{\mathbf{r}}$ is the unit vector defined in Eq. 1.21.
- (c) For the functions in Prob. 1.15, evaluate $(\mathbf{v}_a \cdot \nabla) \mathbf{v}_b$.

- (a) Define $\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$ and $\mathbf{B} = B_x \hat{\mathbf{x}} + B_y \hat{\mathbf{y}} + B_z \hat{\mathbf{z}}$. Then

$$\begin{aligned} (\mathbf{A} \cdot \nabla) \mathbf{B} &= A_x \frac{\partial \mathbf{B}}{\partial x} + A_y \frac{\partial \mathbf{B}}{\partial y} + A_z \frac{\partial \mathbf{B}}{\partial z} \\ &= \sum_{i \in \{x, y, z\}} \left(\sum_{j \in \{x, y, z\}} A_j \frac{\partial B_i}{\partial j} \right) \hat{\mathbf{i}} \end{aligned}$$

We call this expression the **directional derivative**.

- (b) For $i, j \in \{x, y, z\}$, derive

$$\frac{\partial r_i}{\partial j} = \begin{cases} \frac{1}{r} - \frac{j^2}{r^3} & i = j \\ -\frac{ij}{r^3} & i \neq j \end{cases}$$

Then we can compute

$$(\hat{\mathbf{r}} \cdot \nabla) \hat{\mathbf{r}} = \frac{1}{r} \sum_{i \in \{x, y, z\}} \left(\sum_{j \in \{x, y, z\}} j \frac{\partial r_i}{\partial j} \right) \hat{\mathbf{i}}$$

Take $i = x$ to see how this computation works. We have

$$\begin{aligned} [(\hat{\mathbf{r}} \cdot \nabla) \hat{\mathbf{r}}]_x &= \frac{1}{r} \left(x \frac{\partial r_x}{\partial x} + y \frac{\partial r_x}{\partial y} + z \frac{\partial r_x}{\partial z} \right) \\ &= \frac{1}{r} \left[x \left(\frac{1}{r} - \frac{x^2}{r^3} \right) - \frac{xy^2}{r^3} - \frac{xz^2}{r^3} \right] \\ &= \frac{1}{r} \left[x \left(\frac{1}{r} - \left(\frac{x^2 + y^2 + z^2}{r^3} \right) \right) \right] \\ &= \frac{1}{r} \left[x \left(\frac{1}{r} - \frac{1}{r} \right) \right] \\ &= 0 \end{aligned}$$

An extraordinary reduction. By symmetry, it follows that the components for y and z follow analogously. Then we can conclude

$$(\hat{\mathbf{r}} \cdot \nabla) \hat{\mathbf{r}} = \mathbf{0}$$

(c)

$$\begin{aligned}
(\mathbf{v}_a \cdot \nabla) \mathbf{v}_b &= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} v_{a,j} \frac{\partial v_{b,i}}{\partial j} \right) \hat{\mathbf{i}} \\
&= (x^2 y + 3x^2 z^2) \hat{\mathbf{x}} + (6xz^3 - 4xyz) \hat{\mathbf{y}} - 3x^2 z \hat{\mathbf{z}}
\end{aligned}$$

Problem: 1.23

(For masochists only.) Prove product rules (ii) and (vi). Refer to Prob. 1.22 for the definition of $(\mathbf{A} \cdot \nabla) \mathbf{B}$.

$$(ii) \nabla(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A}.$$

PROOF.

$$\begin{aligned}
\nabla(\mathbf{A} \cdot \mathbf{B}) &= \nabla(A_x B_x + A_y B_y + A_z B_z) \\
&= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} A_j \frac{\partial B_j}{\partial i} + B_j \frac{\partial A_j}{\partial i} \right) \hat{\mathbf{i}} \\
\mathbf{A} \times (\nabla \times \mathbf{B}) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ \frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} & \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} & \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \end{vmatrix} \\
&= \left[A_y \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) - A_z \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) \right] \hat{\mathbf{x}} \\
&\quad + \left[A_z \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) - A_x \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) \right] \hat{\mathbf{y}} \\
&\quad + \left[A_x \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) - A_y \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) \right] \hat{\mathbf{z}} \\
\mathbf{B} \times (\nabla \times \mathbf{A}) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ B_x & B_y & B_z \\ \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} & \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} & \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \end{vmatrix} \\
&= \left[B_y \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) - B_z \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \right] \hat{\mathbf{x}} \\
&\quad + \left[B_z \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - B_x \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \right] \hat{\mathbf{y}} \\
&\quad + \left[B_x \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) - B_y \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \right] \hat{\mathbf{z}} \\
(\mathbf{A} \cdot \nabla) \mathbf{B} &= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} A_j \frac{\partial B_i}{\partial j} \right) \hat{\mathbf{i}} \\
(\mathbf{B} \cdot \nabla) \mathbf{A} &= \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} B_j \frac{\partial A_i}{\partial j} \right) \hat{\mathbf{i}}
\end{aligned}$$

Adding $\mathbf{A} \times (\nabla \times \mathbf{B})$ and $\mathbf{B} \times (\nabla \times \mathbf{A})$, derive

$$\sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} (1 - \delta_{ij}) \left(A_j \frac{\partial B_j}{\partial i} + B_j \frac{\partial A_j}{\partial i} \right) \right) \hat{\mathbf{i}} \\ - \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} (1 - \delta_{ij}) \left(A_j \frac{\partial B_i}{\partial j} + B_j \frac{\partial A_i}{\partial j} \right) \right) \hat{\mathbf{i}}$$

Now, looking at the directional derivatives, we may write their sum $(\mathbf{A} \cdot \nabla) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{A}$ as

$$\sum_{i \in \{x,y,z\}} \left(A_i \frac{\partial B_i}{\partial i} + B_i \frac{\partial A_i}{\partial i} \right) \hat{\mathbf{i}} + \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} (1 - \delta_{ij}) \left(A_j \frac{\partial B_i}{\partial j} + B_j \frac{\partial A_i}{\partial j} \right) \right) \hat{\mathbf{i}}$$

The blue summations cancel each other out, while the red terms sum to the desired result. Observe that the heart of the proof arose in decomposing the derivative structures: the first corresponds to the gradient and the second to the directional derivative.

$$A_j \frac{\partial B_j}{\partial i} \text{ (gradient)} \quad A_j \frac{\partial B_i}{\partial j} \text{ (directional derivative)}$$

□

$$(vi) \nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}).$$

PROOF.

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_y B_z - A_z B_y & A_z B_x - A_x B_z & A_x B_y - A_y B_x \end{vmatrix} \\ = \sum_{i \in \{x,y,z\}} \left\{ \sum_{j \in \{x,y,z\}} \left[\left(A_i \frac{\partial B_j}{\partial j} + B_j \frac{\partial A_i}{\partial j} \right) - \left(A_j \frac{\partial B_i}{\partial j} + B_i \frac{\partial A_j}{\partial j} \right) \right] \right\} \hat{\mathbf{i}} \\ (\mathbf{B} \cdot \nabla) \mathbf{A} = \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} B_j \frac{\partial A_i}{\partial j} \right) \hat{\mathbf{i}} \\ (\mathbf{A} \cdot \nabla) \mathbf{B} = \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} A_j \frac{\partial B_i}{\partial j} \right) \hat{\mathbf{i}} \\ \mathbf{A}(\nabla \cdot \mathbf{B}) = \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} A_i \frac{\partial B_j}{\partial j} \right) \hat{\mathbf{i}} \\ \mathbf{B}(\nabla \cdot \mathbf{A}) = \sum_{i \in \{x,y,z\}} \left(\sum_{j \in \{x,y,z\}} B_i \frac{\partial A_j}{\partial j} \right) \hat{\mathbf{i}}$$

Where once again, we combine **divergence** and **directional derivative** components to produce the desired result. □

Problem: 1.24

Derive the three quotient rules.

Switch either $g \rightarrow g^{-1}$ or $f \rightarrow f^{-1}$ to find

$$\begin{aligned}
 \nabla \left(\frac{f}{g} \right) &= \nabla (f g^{-1}) \\
 &= f \nabla g^{-1} + g^{-1} \nabla f \\
 &= \frac{g \nabla f - f \nabla g}{g^2} \\
 \nabla \cdot \left(\frac{1}{f} \mathbf{A} \right) &= \nabla \cdot (f^{-1} \mathbf{A}) \\
 &= f^{-1} (\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f^{-1}) \\
 &= \frac{f (\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot \nabla f}{f^2} \\
 \nabla \times \left(\frac{1}{f} \mathbf{A} \right) &= \nabla \times (f^{-1} \mathbf{A}) \\
 &= f^{-1} (\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f^{-1}) \\
 &= \frac{f (\nabla \times \mathbf{A}) + \mathbf{A} \times \nabla f}{f^2}
 \end{aligned}$$

Problem: 1.25

- (a) Check product rule (iv) (by calculating each term separately) for the functions

$$\mathbf{A} = x\hat{\mathbf{x}} + 2y\hat{\mathbf{y}} + 3z\hat{\mathbf{z}} \quad \mathbf{B} = 3y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}$$

- (b) Do the same for product rule (ii).
(c) Do the same for rule (vi).

Use the fact that $\nabla \times \mathbf{A} = \mathbf{0}$ from our insight in problem 1.20 that a zero curl arises when each component function is strictly in the variable of its corresponding basis coordinate.

(a)

$$\begin{aligned}
\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ x & 2y & 3z \\ 3y & -2x & 0 \end{vmatrix} \\
&= 6xz\hat{\mathbf{x}} + 9yz\hat{\mathbf{y}} - (2x^2 + 6y^2)\hat{\mathbf{z}} \\
\boxed{\nabla \cdot (\mathbf{A} \times \mathbf{B})} &= \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(9yz) - \frac{\partial}{\partial z}(2x^2 + 6y^2) \\
&= \boxed{15z} \\
\boxed{\mathbf{B} \cdot (\nabla \times \mathbf{A})} &= \boxed{0} \\
\nabla \times \mathbf{B} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 3y & -2x & 0 \end{vmatrix} \\
&= -5\hat{\mathbf{z}} \\
\boxed{\mathbf{A} \cdot (\nabla \times \mathbf{B})} &= \boxed{-15z} \\
\boxed{\mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})} &= \boxed{15z}
\end{aligned}$$

(b)

$$\begin{aligned}
\boxed{\nabla(\mathbf{A} \cdot \mathbf{B})} &= \frac{\partial}{\partial x}(-xy)\hat{\mathbf{x}} + \frac{\partial}{\partial y}(-xy)\hat{\mathbf{y}} \\
&= \boxed{-y\hat{\mathbf{x}} - x\hat{\mathbf{y}}} \\
\nabla \times \mathbf{B} &= -5\hat{\mathbf{z}} \\
\boxed{\mathbf{A} \times (\nabla \times \mathbf{B})} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ x & 2y & 3z \\ 0 & 0 & -5 \end{vmatrix} \\
&= \boxed{-10y\hat{\mathbf{x}} + 5x\hat{\mathbf{y}}} \\
\boxed{\mathbf{B} \times (\nabla \times \mathbf{A})} &= \boxed{\mathbf{0}} \\
\boxed{(\mathbf{A} \cdot \nabla)\mathbf{B}} &= x\frac{\partial \mathbf{B}}{\partial x} + 2y\frac{\partial \mathbf{B}}{\partial y} \\
&= \boxed{6y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}} \\
\boxed{(\mathbf{B} \cdot \nabla)\mathbf{A}} &= 3y\frac{\partial \mathbf{A}}{\partial x} - 2x\frac{\partial \mathbf{A}}{\partial y} \\
&= \boxed{3y\hat{\mathbf{x}} - 4x\hat{\mathbf{y}}}
\end{aligned}$$

Summing our terms together, derive

$$\mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} = \boxed{-y\hat{\mathbf{x}} - x\hat{\mathbf{y}}}$$

(c)

$$\begin{aligned}
\boxed{\nabla \times (\mathbf{A} \times \mathbf{B})} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xz & 9yz & -(2x^2 + 6y^2) \end{vmatrix} \\
&= \left[\frac{\partial}{\partial y} [-(2x^2 + 6y^2)] - \frac{\partial}{\partial z} (9yz) \right] \hat{\mathbf{x}} \\
&\quad + \left[\frac{\partial}{\partial z} (6xz) + \frac{\partial}{\partial x} (2x^2 + 6y^2) \right] \hat{\mathbf{y}} \\
&\quad + \left[\frac{\partial}{\partial x} (9yz) - \frac{\partial}{\partial y} (6xz) \right] \hat{\mathbf{z}} \\
&= \boxed{-21y\hat{\mathbf{x}} + 10x\hat{\mathbf{y}}} \\
\boxed{(\mathbf{B} \cdot \nabla) \mathbf{A}} &= \boxed{3y\hat{\mathbf{x}} - 4x\hat{\mathbf{y}}} \\
\boxed{(\mathbf{A} \cdot \nabla) \mathbf{B}} &= \boxed{6y\hat{\mathbf{x}} - 2x\hat{\mathbf{y}}} \\
\boxed{\mathbf{A}(\nabla \cdot \mathbf{B})} &= \boxed{\mathbf{0}} \\
\boxed{\mathbf{B}(\nabla \cdot \mathbf{A})} &= \boxed{6\mathbf{B}} \\
&= \boxed{18y\hat{\mathbf{x}} - 12x\hat{\mathbf{y}}}
\end{aligned}$$

Combining the last four terms as rule (vi) stipulates, we have

$$(\mathbf{B} \cdot \nabla) \mathbf{A} - (\mathbf{A} \cdot \nabla) \mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) = -21y\hat{\mathbf{x}} + 10x\hat{\mathbf{y}}$$

1.2.7 - Second Derivatives

Problem: 1.26

Calculate the Laplacian of the following functions:

- (a) $T_a = x^2 + 2xy + 3z + 4$
- (b) $T_b = \sin(x) \sin(y) \sin(z)$
- (c) $T_c = e^{-5x} \sin(4y) \cos(3z)$
- (d) $\mathbf{v} = x^2\hat{\mathbf{x}} + 3xz^2\hat{\mathbf{y}} - 2xz\hat{\mathbf{z}}$

(a)

$$\nabla^2 T_a = \frac{\partial^2 T_a}{\partial x^2} + \frac{\partial^2 T_a}{\partial y^2} + \frac{\partial^2 T_a}{\partial z^2} = 2$$

(b)

$$\nabla^2 T_b = \frac{\partial^2 T_b}{\partial x^2} + \frac{\partial^2 T_b}{\partial y^2} + \frac{\partial^2 T_b}{\partial z^2} = -3 \sin(x) \sin(y) \sin(z)$$

(c)

$$\nabla^2 T_c = \frac{\partial^2 T_c}{\partial x^2} + \frac{\partial^2 T_c}{\partial y^2} + \frac{\partial^2 T_c}{\partial z^2} = (25 - 16 - 3) e^{-5x} \sin(4y) \cos(3z) = 0$$

(d)

$$\nabla^2 \mathbf{v} = \nabla^2(x^2) \hat{\mathbf{x}} + \nabla^2(3xz^2) \hat{\mathbf{y}} + \nabla^2(-2xz) \hat{\mathbf{z}} = 2\hat{\mathbf{x}} + 6xz\hat{\mathbf{y}} - 2\hat{\mathbf{z}}$$

Problem: 1.27

Prove that the divergence of a curl is always zero. *Check* it for function $\mathbf{v}_a$ in Prob. 1.15.

$$\nabla \cdot (\nabla \times \mathbf{v}) = 0.$$

PROOF.

$$\begin{aligned} \nabla \times \mathbf{v} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} \\ &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \hat{\mathbf{z}} \\ \nabla \cdot (\nabla \times \mathbf{v}) &= \frac{\partial}{\partial x} \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \end{aligned}$$

By Clairaut's theorem, assuming continuity of the mixed second partial derivatives, the differentiation is agnostic of order:

$$\frac{\partial^2 v_i}{\partial j \partial k} = \frac{\partial^2 v_i}{\partial k \partial j}$$

where the three colored pairs (after distribution) cancel pursuant to Clairaut's theorem. $\square$

Using $\mathbf{v} = \mathbf{v}_a$, we have

$$\begin{aligned} \nabla \times \mathbf{v}_a &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 3xz^2 & -2xz \end{vmatrix} \\ &= -6xz\hat{\mathbf{x}} + 2z\hat{\mathbf{y}} + 3z^2\hat{\mathbf{z}} \\ \nabla \cdot (\nabla \times \mathbf{v}_a) &= \frac{\partial}{\partial x}(-6xz) + \frac{\partial}{\partial y}(2z) + \frac{\partial}{\partial z}(3z^2) \\ &= 0 \end{aligned}$$

Problem: 1.28

Prove that the curl of a gradient is always zero. *Check* it for function (b) in Prob. 1.11.

$$\nabla \times (\nabla T) = \mathbf{0}.$$

PROOF.

$$\begin{aligned}
 \nabla \times (\nabla T) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial T}{\partial x} & \frac{\partial T}{\partial y} & \frac{\partial T}{\partial z} \end{vmatrix} \\
 &= \left(\frac{\partial^2 T}{\partial y \partial z} - \frac{\partial^2 T}{\partial z \partial y} \right) \hat{\mathbf{x}} + \left(\frac{\partial^2 T}{\partial z \partial x} - \frac{\partial^2 T}{\partial x \partial z} \right) \hat{\mathbf{y}} + \left(\frac{\partial^2 T}{\partial x \partial y} - \frac{\partial^2 T}{\partial y \partial x} \right) \hat{\mathbf{z}} \\
 &= \mathbf{0}
 \end{aligned}$$

where the last equality follows from Clairaut's theorem. $\square$

Using $f(x, y, z) = x^2 y^3 z^4$, calculate

$$\begin{aligned}
 \nabla f &= 2xy^3z^4 \hat{\mathbf{x}} + 3x^2y^2z^4 \hat{\mathbf{y}} + 4x^2y^3z^3 \hat{\mathbf{z}} \\
 \nabla \times (\nabla f) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy^3z^4 & 3x^2y^2z^4 & 4x^2y^3z^3 \end{vmatrix} \\
 &= \left[\frac{\partial}{\partial y} (4x^2y^3z^3) - \frac{\partial}{\partial z} (3x^2y^2z^4) \right] \hat{\mathbf{x}} \\
 &\quad + \left[\frac{\partial}{\partial z} (2xy^3z^4) - \frac{\partial}{\partial x} (4x^2y^3z^3) \right] \hat{\mathbf{y}} \\
 &\quad + \left[\frac{\partial}{\partial x} (3x^2y^2z^4) - \frac{\partial}{\partial y} (2xy^3z^4) \right] \hat{\mathbf{z}} \\
 &= \mathbf{0}
 \end{aligned}$$

1.3 - Integral Calculus

1.3.1 - Line, Surface, and Volume Integrals

Problem: 1.29

Calculate the line integral of the function $\mathbf{v} = x^2\hat{\mathbf{x}} + 2yz\hat{\mathbf{y}} + y^2\hat{\mathbf{z}}$ from the origin to the point $(1, 1, 1)$ by three different routes:

- (a) $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1)$.
- (b) $(0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1)$.
- (c) The direct straight line.
- (d) What is the line integral around the closed loop that goes *out* along path (a) and *back* along path (b)?

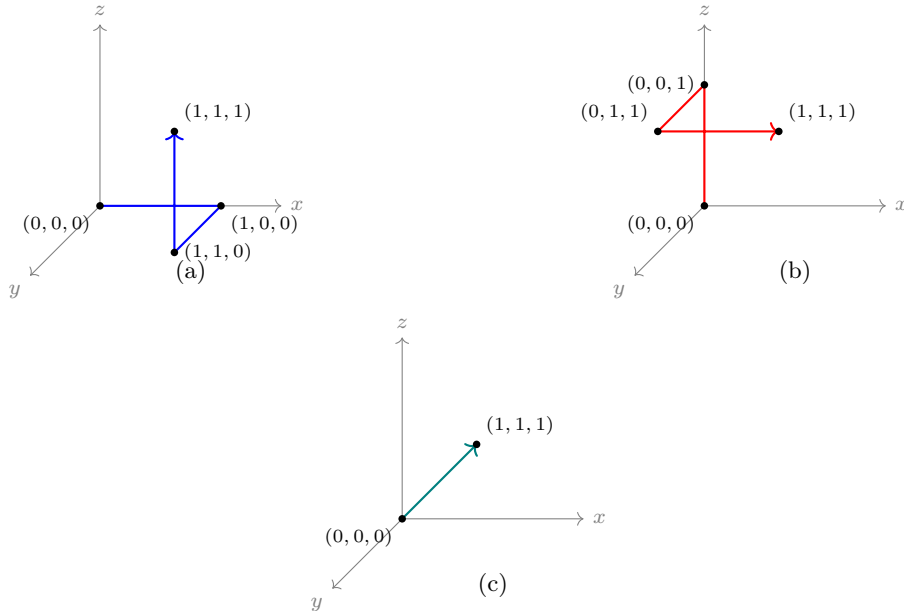


Figure 1: The three integration routes of Problem 1.29.

For all parts, $d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$.

- (a) $(0, 0, 0) \rightarrow (1, 0, 0)$: We have $dy = dz = 0$ and $y = z = 0$. Calculate

$$\int_0^1 x^2 dx = \frac{1}{3}$$

$(1, 0, 0) \rightarrow (1, 1, 0)$: We have $dx = dz = 0$, $x = 1$, and $z = 0$. Then $\mathbf{v} \cdot d\mathbf{l} = 0$, leaving us with 0 contribution on this section.

$(1, 1, 0) \rightarrow (1, 1, 1)$: We have $dx = dy = 0$ and $x = y = 1$. Calculate

$$\int_0^1 dz = 1$$

Adding the contributions from each path produces $4/3$.

(b) $(0, 0, 0) \rightarrow (0, 0, 1)$: We have $dx = dy = 0$ and $x = y = 0$. Since $\mathbf{v} \cdot d\mathbf{l} = 0$, the contribution on this section is 0.

$(0, 0, 1) \rightarrow (0, 1, 1)$: We have $dx = dz = 0$, $x = 0$, and $z = 1$. Calculate

$$\int_0^1 2y dy = 1$$

$(0, 1, 1) \rightarrow (1, 1, 1)$: We have $dy = dz = 0$ and $y = z = 1$. Calculate

$$\int_0^1 x^2 dx = \frac{1}{3}$$

Summing each section's contribution, we have $4/3$ yet again.

(c) Since the line connecting $(0, 0, 0)$ and $(1, 1, 1)$ is the set of all (x, y, z) such that $x = y = z$, parameterize as t . Then $dx = dy = dz = dt$, $d\mathbf{l} = dt(\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}})$, and $\mathbf{v} = t^2\hat{\mathbf{x}} + 2t^2\hat{\mathbf{y}} + t^2\hat{\mathbf{z}}$. Thus $\mathbf{v} \cdot d\mathbf{l}$ gives us

$$\mathbf{v} \cdot d\mathbf{l} = 4t^2 dt$$

Integrating over $t \in [0, 1]$, compute

$$\int_0^1 4t^2 dt = \frac{4}{3}$$

(d)

$$\oint \mathbf{v} \cdot d\mathbf{l} = \frac{4}{3} - \frac{4}{3} = 0$$

These answers are no coincidence, for later we will learn that a vector field $\mathbf{v}$ with the property

$$\nabla \times \mathbf{v} = \mathbf{0}$$

is a **conservative** force. As it turns out, one can verify that this is the case for our $\mathbf{v}$.

Problem: 1.30

Calculate the surface integral of the function in Ex. 1.7, over the *bottom* of the box. For consistency, let “upward” be the positive direction. Does the surface integral depend only on the boundary line for this function? What is the total flux over the *closed* surface of the box (*including* the bottom)? [Note: For the *closed* surface, the positive direction is “outward,” and hence “down,” for the bottom face.]

We have $z = 0$, $d\mathbf{a} = -dx\,dy\,\hat{\mathbf{z}}$, and $\mathbf{v} \cdot d\mathbf{a} = 3y\,dx\,dy$, so we compute

$$\int_0^2 dx \int_0^2 3y\,dy = 12$$

From the flux of the other five sides, we sum: $20 + 12 = 32$. In this case, the surface integral depends on the surface itself. The boundary line in equation is the square spanned by $(0, 0, 0)$ and $(2, 2, 0)$, which is the boundary line shared by the bottom face, or by the total surface of the *other* five faces combined. Since each surface has a different flux, the surface integral is surface-dependent.

Problem: 1.31

Calculate the volume integral of the function $T = z^2$ over the tetrahedron with corners at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$.

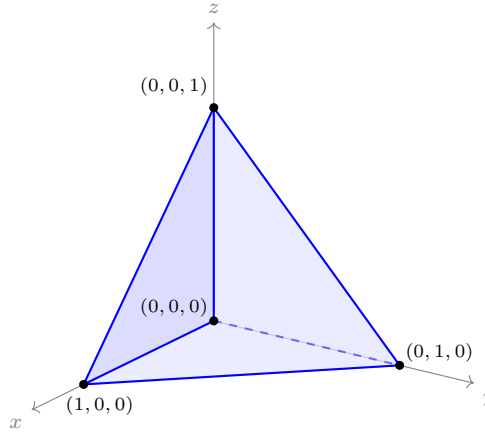


Figure 2: The tetrahedron of Problem 1.31, with vertices at the origin and the three unit axis points.

The plane traced out by the tetrahedron has shape $x + y + z = 1$. A clever approach is to consider a horizontal cross-section of the tetrahedron (i.e. fixing z at some constant), and the dimensions of the triangle traced out by that level surface:

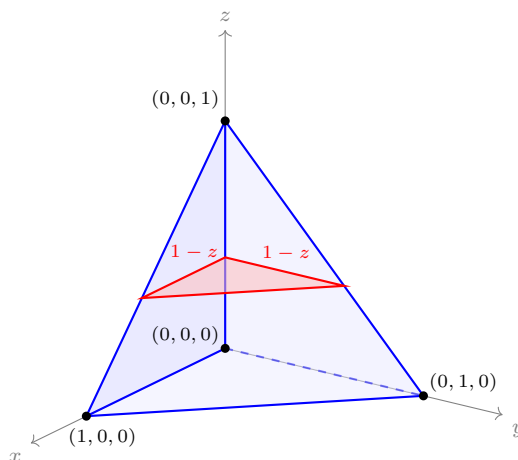


Figure 3: Horizontal cross-section of the tetrahedron at height z : a right triangle with legs $1 - z$ and area $\frac{1}{2}(1 - z)^2$.

Upon fixing the height of that level surface, we can see that the right triangle traced out by the level surface intersecting with the slanted face of the tetrahedron has legs of equal length $1 - z$. The area of this triangle is $\frac{1}{2}(1 - z)^2$; critically, we ought to make the observation that

$$\frac{1}{2}(1 - z)^2 = dx \, dy$$

This simplifies our volume integral to

$$\int_0^1 z^2 \cdot \frac{1}{2}(1 - z)^2 \, dz = \frac{1}{2} \int_0^1 (z^2 - 2z^3 + z^4) \, dz = \frac{1}{2} \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right) = \frac{1}{60}$$

1.3.3 - The Fundamental Theorem of Gradients

Problem: 1.32

Check the fundamental theorem for gradients, using $T = x^2 + 4xy + 2yz^3$, the points $\mathbf{a} = (0, 0, 0)$, $\mathbf{b} = (1, 1, 1)$, and the three paths in Fig. 1.28:

- (a) $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1)$;
- (b) $(0, 0, 0) \rightarrow (0, 0, 1) \rightarrow (0, 1, 1) \rightarrow (1, 1, 1)$;
- (c) the parabolic path $z = x^2$; $y = x$.

The gradient is

$$\nabla T = (2x + 4y) \hat{\mathbf{x}} + (4x + 2z^3) \hat{\mathbf{y}} + 6yz^2 \hat{\mathbf{z}}$$

- (a) (i) $(0, 0, 0) \rightarrow (1, 0, 0)$:

$$y = z = dy = dz = 0$$

$$\nabla T \cdot d\mathbf{l} = 2x \, dx$$

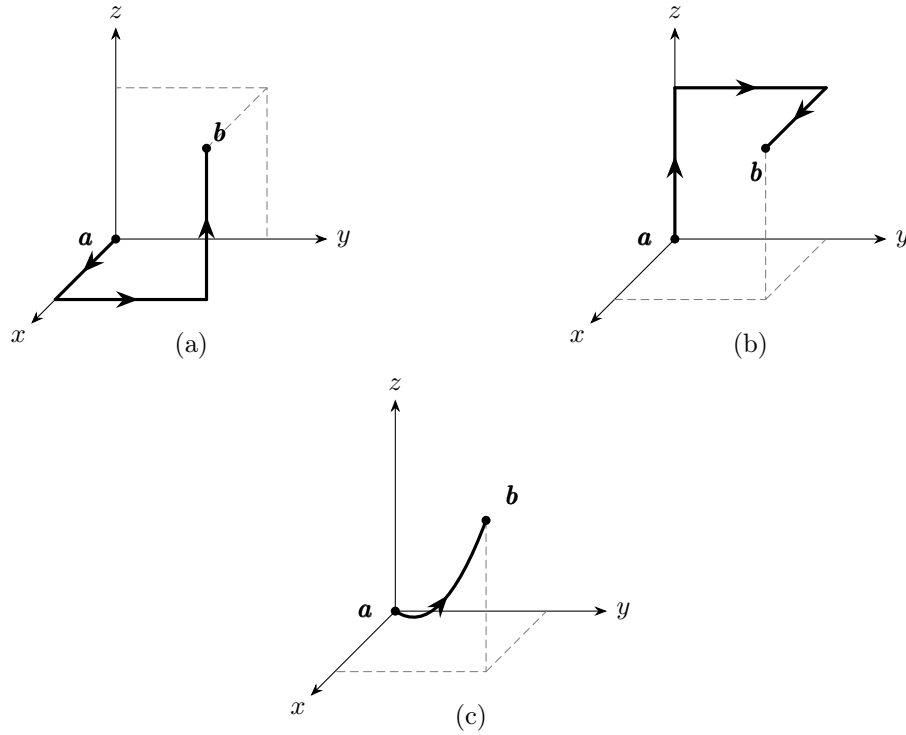


Figure 4: The three paths of Problem 1.32 (cf. Griffiths Fig. 1.28).

$$\int_0^1 2x \, dx = 1$$

(ii) $(1, 0, 0) \rightarrow (1, 1, 0)$:

$$x = 1$$

$$z = dx = dz = 0$$

$$\nabla T \cdot d\mathbf{l} = (4x + 2z^3) \, dy$$

$$\int_0^1 4 \, dy = 4$$

(iii) $(1, 1, 0) \rightarrow (1, 1, 1)$:

$$x = y = 1$$

$$dx = dy = 0$$

$$\nabla T \cdot d\mathbf{l} = 6z^2 \, dz$$

$$\int_0^1 6z^2 \, dz = 2$$

Combining the path contributions,

$$\int_a^b \nabla T \cdot d\mathbf{l} = 7$$

(b) (i) $(0, 0, 0) \rightarrow (0, 0, 1)$:

$$x = y = dx = dy = 0$$

$$\nabla T \cdot d\mathbf{l} = 0$$

$$\int_0^1 0 \, dz = 0$$

(ii) $(0, 0, 1) \rightarrow (0, 1, 1)$:

$$x = dx = dz = 0$$

$$z = 1$$

$$\nabla T \cdot d\mathbf{l} = 2 \, dy$$

$$\int_0^1 2 \, dy = 2$$

(iii) $(0, 1, 1) \rightarrow (1, 1, 1)$:

$$y = z = 1$$

$$dy = dz = 0$$

$$\nabla T \cdot d\mathbf{l} = (2x + 4) \, dx$$

$$\int_0^1 (2x + 4) \, dx = 5$$

Combining the path contributions, we derive the same result, as anticipated from the fundamental theorem for gradients.

$$\int_a^b \nabla T \cdot d\mathbf{l} = 7$$

(c) From the parabola definition, derive

$$dz = 2x \, dx, \quad dy = dx$$

Then the infinitesimal displacement becomes

$$d\mathbf{l} = (\hat{\mathbf{x}} + \hat{\mathbf{y}} + 2x\hat{\mathbf{z}}) \, dx$$

and the gradient, now written exclusively in x :

$$\nabla T = 6x\hat{\mathbf{x}} + (4x + 2x^6)\hat{\mathbf{y}} + 6x^5\hat{\mathbf{z}}$$

Compute the dot product as

$$\nabla T \cdot d\mathbf{l} = 10x + 14x^6$$

Integrating the line integral, compute

$$\int_0^1 (10x + 14x^6) \, dx = (5x^2 + 2x^7) \Big|_0^1 = 7$$

As expected, the fundamental theorem of gradient works!

1.3.4 - The Fundamental Theorem for Divergences

Problem: 1.33

Test the divergence theorem for the function $\mathbf{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3zx)\hat{\mathbf{z}}$. Take as your volume the cube shown in Fig. 1.30, with sides of length 2.

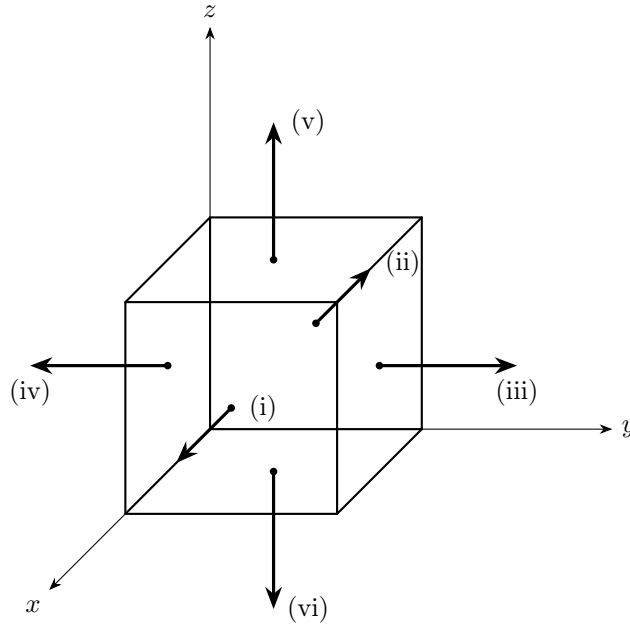


Figure 5: Cube of length 2 with outward normals on each face (cf. Griffiths Fig. 1.30).

The divergence is

$$\nabla \cdot \mathbf{v} = 3x + y + 2z$$

Then its volume integral is

$$\int_V \nabla \cdot \mathbf{v} \, d\tau = \int_0^2 \int_0^2 \int_0^2 (3x + y + 2z) \, dx \, dy \, dz = 48$$

Now for the surface integral:

$$\oint_S \mathbf{v} \cdot d\mathbf{a}$$

(i) We have $x = 2$ and $\mathbf{v} \cdot dy \, dz \hat{\mathbf{x}} = 2y \, dy \, dz$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 \int_0^2 2y \, dy \, dz = 8$$

(ii) $x = 0$ and $-\mathbf{v} \cdot dy \, dz \hat{\mathbf{x}} = 0 \, dy \, dz$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 \int_0^2 0 \, dy \, dz = 0$$

(iii) $y = 2$ and $\mathbf{v} \cdot d\mathbf{x} dz \hat{\mathbf{y}} = 4z dx dz$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 \int_0^2 4z dx dz = 16$$

(iv) $y = 0$ and $-\mathbf{v} \cdot d\mathbf{x} dz \hat{\mathbf{y}} = 0 dx dz$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 \int_0^2 0 dx dz = 0$$

(v) $z = 2$ and $\mathbf{v} \cdot dx dy \hat{\mathbf{z}} = 6x dx dy$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 \int_0^2 6x dx dy = 24$$

(vi) $z = 0$ and $-\mathbf{v} \cdot dx dy \hat{\mathbf{z}} = 0 dx dy$.

$$\int \mathbf{v} \cdot d\mathbf{a} = \int_0^2 \int_0^2 0 dx dy = 0$$

Summing the fluxes produces

$$\oint_S \mathbf{v} \cdot d\mathbf{a} = 8 + 16 + 24 = 48$$

verifying the divergence theorem.

1.3.5 - The Fundamental Theorem for Curls

Problem: 1.34

Test Stokes' theorem for the function $\mathbf{v} = (xy)\hat{\mathbf{x}} + (2yz)\hat{\mathbf{y}} + (3zx)\hat{\mathbf{z}}$ using the triangular shaded area of Fig. 1.34.

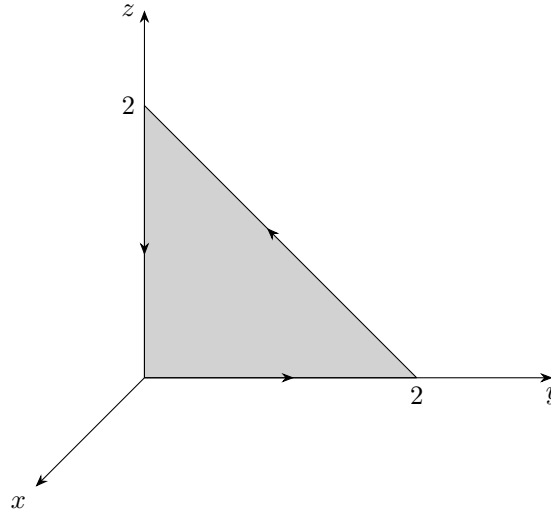


Figure 6: Griffiths Figure 1.34

Compute

$$\begin{aligned}\nabla \times \mathbf{v} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & 2yz & 3zx \end{vmatrix} \\ &= -2y\hat{\mathbf{x}} - 3z\hat{\mathbf{y}} - x\hat{\mathbf{z}}\end{aligned}$$

The surface integral is

$$\begin{aligned}\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} &= - \int_0^2 \int_0^{2-z} 2y \, dy \, dz \\ &= - \int_0^2 (2-z)^2 \, dz \\ &= -4z + 2z^2 - \frac{z^3}{3} \Big|_0^2 \\ &= -\frac{8}{3}\end{aligned}$$

Now for the line integral:

(i) $(0, 0, 0) \rightarrow (0, 2, 0)$: $x = z = dx = dz = 0$, $d\mathbf{l} = dy\hat{\mathbf{y}}$, $\mathbf{v} \cdot d\mathbf{l} = 0 \, dy$

$$\int \mathbf{v} \cdot d\mathbf{l} = 0$$

(ii) $(0, 2, 0) \rightarrow (0, 0, 2)$:

$$x = dx = 0$$

$$d\mathbf{l} = dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}} = (\hat{\mathbf{y}} - \hat{\mathbf{z}}) dy$$

$$\mathbf{v} \cdot d\mathbf{l} = 2yz dy = 2y(2 - y) dy$$

where the hypotenuse of the triangle has equation $z = 2 - y$, producing $dz/dy = -1 \iff dz = -dy$. This segment's contribution is

$$\int \mathbf{v} \cdot d\mathbf{l} = \int_2^0 2y(2 - y) dy = \left. \frac{2}{3}y^3 - 2y^2 \right|_0^2 = -\frac{8}{3}$$

where the bounds are flipped to reflect the movement along the hypotenuse in the negative y -direction.

(iii) $(0, 0, 2) \rightarrow (0, 0, 0)$: $x = y = dx = dy = 0$, $d\mathbf{l} = dz\hat{\mathbf{z}}$, $\mathbf{v} \cdot d\mathbf{l} = 0 dz$

$$\int \mathbf{v} \cdot d\mathbf{l} = 0$$

Summed together, we verify Stokes' theorem:

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \oint \mathbf{v} \cdot d\mathbf{l} = -\frac{8}{3}$$

Problem: 1.35

Check Corollary 1 by using the same function and boundary line as in Ex. 1.11, but integrating over the five faces of the cube in Fig. 1.35. The back of the cube is open.

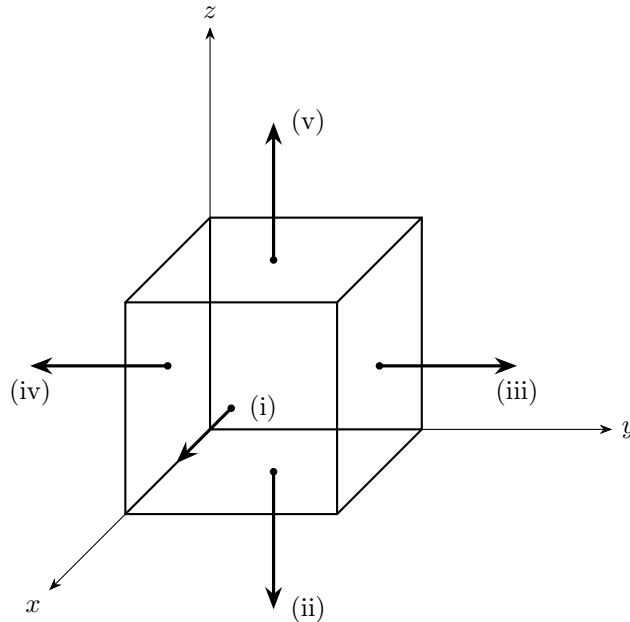


Figure 7: Unit cube with outward normals on each face (cf. Griffiths Fig. 1.35).

Our function is $\mathbf{v} = (2xz + 3y^2) \hat{\mathbf{y}} + (4yz^2) \hat{\mathbf{z}}$, with

$$\nabla \times \mathbf{v} = (4z^2 - 2x) \hat{\mathbf{x}} + 2z \hat{\mathbf{z}}$$

computed in Ex. 1.11. Corollary 1 states that $\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a}$ depends only on the boundary line, so integrating over the five faces should reproduce the value $\frac{4}{3}$ obtained in Ex. 1.11 for the square surface at $x = 0$. The right-hand rule applied to the boundary circulation makes the normals point outward.

(i) $x = 1$, $d\mathbf{a} = dy dz \hat{\mathbf{x}}$, $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = (4z^2 - 2) dy dz$.

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^1 \int_0^1 (4z^2 - 2) dy dz = \frac{4}{3} - 2 = -\frac{2}{3}$$

(ii) $z = 0$, $d\mathbf{a} = -dx dy \hat{\mathbf{z}}$, $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = -2z dx dy = 0 dx dy$.

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$$

(iii) $y = 1$, $d\mathbf{a} = dx dz \hat{\mathbf{y}}$, and $\nabla \times \mathbf{v}$ has no $\hat{\mathbf{y}}$ component.

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$$

(iv) $y = 0$, $d\mathbf{a} = -dx dz \hat{\mathbf{y}}$, likewise.

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$$

(v) $z = 1$, $d\mathbf{a} = dx dy \hat{\mathbf{z}}$, $(\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 2z dx dy = 2 dx dy$.

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^1 \int_0^1 2 dx dy = 2$$

Summing the fluxes produces

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = -\frac{2}{3} + 2 = \frac{4}{3}$$

matching Ex. 1.11 and confirming Corollary 1.

1.3.6 - Integration by Parts

Problem: 1.36

(a) Show that

$$\int_S f(\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \int_S [\mathbf{A} \times (\nabla f)] \cdot d\mathbf{a} + \oint_P f \mathbf{A} \cdot d\mathbf{l}$$

(b) Show that

$$\int_V \mathbf{B} \cdot (\nabla \times \mathbf{A}) d\tau = \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) d\tau + \oint_S (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a}$$

(a) From the product rule

$$\nabla \times (f\mathbf{A}) = f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f)$$

Rewrite as

$$f(\nabla \times \mathbf{A}) = \mathbf{A} \times (\nabla f) + \nabla \times (f\mathbf{A})$$

Take the surface integral of each term to find

$$\int_S f(\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \int_S [\mathbf{A} \times (\nabla f)] \cdot d\mathbf{a} + \int_S \nabla \times (f\mathbf{A}) \cdot d\mathbf{a}$$

On the last term, apply Stokes' theorem to convert to a loop integral, leaving us with

$$\int_S f(\nabla \times \mathbf{A}) \cdot d\mathbf{a} = \int_S [\mathbf{A} \times (\nabla f)] \cdot d\mathbf{a} + \oint_P f \mathbf{A} \cdot d\mathbf{l}$$

(b) From the product rule

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

Rewrite as

$$\mathbf{B} \cdot (\nabla \times \mathbf{A}) = \mathbf{A} \cdot (\nabla \times \mathbf{B}) + \nabla \cdot (\mathbf{A} \times \mathbf{B})$$

Take the volume integral of each term to find

$$\int_V \mathbf{B} \cdot (\nabla \times \mathbf{A}) d\tau = \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) d\tau + \int_V \nabla \cdot (\mathbf{A} \times \mathbf{B}) d\tau$$

By the divergence theorem, convert the last term to the surface flux to derive

$$\int_V \mathbf{B} \cdot (\nabla \times \mathbf{A}) d\tau = \int_V \mathbf{A} \cdot (\nabla \times \mathbf{B}) d\tau + \oint_S (\mathbf{A} \times \mathbf{B}) \cdot d\mathbf{a}$$

1.4 - Curvilinear Coordinates

1.4.1 - Spherical Coordinates

Problem: 1.37

Find formulas for r , θ , ϕ in terms of x , y , z (the inverse, in other words, of Eq. 1.62).

The transformation from Cartesian to spherical coordinates is given by Equation 1.62:

$$x = r \sin(\theta) \cos(\phi), \quad y = r \sin(\theta) \sin(\phi), \quad z = r \cos(\theta)$$

The radial length is derived as the positive root of

$$\begin{aligned} r^2 &= r^2 [\sin^2(\theta) + \cos^2(\theta)] \\ &= [r^2 \sin^2(\theta) \cos^2(\phi) + r^2 \sin^2(\theta) \sin^2(\phi)] + r^2 \cos^2(\theta) \\ &= x^2 + y^2 + z^2 \end{aligned}$$

For θ , in the z -equation, isolate θ :

$$\frac{z}{r} = \frac{z}{\sqrt{x^2 + y^2 + z^2}} = \cos(\theta) \quad \implies \quad \theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

Lastly, for ϕ , divide y by x to find

$$\frac{y}{x} = \tan(\phi) \quad \implies \quad \phi = \tan^{-1} \left(\frac{y}{x} \right)$$

which is specifically defined on $(-\pi/2, \pi/2)$.

Problem: 1.38

Express the unit vectors $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$ in terms of $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ (that is, derive Eq. 1.64). Check your answers several ways ($\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} \stackrel{?}{=} 1$, $\hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} \stackrel{?}{=} 0$, $\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} \stackrel{?}{=} \hat{\boldsymbol{\phi}}$...). Also work out the inverse formulas, giving $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ in terms of $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, $\hat{\boldsymbol{\phi}}$ (and θ , ϕ).

Using the mappings from (x, y, z) to (r, θ, ϕ) in Equation 1.62, we can first derive our unit radial vector as

$$\hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|} = \sin(\theta) \cos(\phi) \hat{\mathbf{x}} + \sin(\theta) \sin(\phi) \hat{\mathbf{y}} + \cos(\theta) \hat{\mathbf{z}}$$

where $|\mathbf{r}| = r$. To find the direction of increasing θ , which is definitionally what $\hat{\boldsymbol{\theta}}$ is, we first differentiate $\hat{\mathbf{r}}$ with respect to θ to get $\boldsymbol{\theta} = \partial \hat{\mathbf{r}} / \partial \theta$, and normalize to find

$$\hat{\boldsymbol{\theta}} = \frac{\partial \hat{\mathbf{r}} / \partial \theta}{|\partial \hat{\mathbf{r}} / \partial \theta|} = \frac{\boldsymbol{\theta}}{|\boldsymbol{\theta}|} = \cos(\theta) \cos(\phi) \hat{\mathbf{x}} + \cos(\theta) \sin(\phi) \hat{\mathbf{y}} - \sin(\theta) \hat{\mathbf{z}}$$

where $|\boldsymbol{\theta}| = 1$. Lastly, differentiate $\hat{\mathbf{r}}$ with respect to ϕ to get $\boldsymbol{\phi} = \partial\hat{\mathbf{r}}/\partial\phi$, and normalize:

$$\hat{\boldsymbol{\phi}} = \frac{\partial\hat{\mathbf{r}}/\partial\phi}{|\partial\hat{\mathbf{r}}/\partial\phi|} = \frac{\boldsymbol{\phi}}{|\boldsymbol{\phi}|} = -\sin(\phi)\hat{\mathbf{x}} + \cos(\phi)\hat{\mathbf{y}}$$

where $|\boldsymbol{\phi}| = \sin(\theta)$. Verifying in the recommended ways, we can compute

$$\begin{aligned}\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} &= \sin^2(\theta) [\cos^2(\phi) + \sin^2(\phi)] + \cos^2(\theta) \\ &= \sin^2(\theta) + \cos^2(\theta) \\ &= 1 \\ \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} &= -\cos(\theta)\cos(\phi)\sin(\phi) + \cos(\theta)\sin(\phi)\cos(\phi) \\ &= 0 \\ \hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \sin(\theta)\cos(\phi) & \sin(\theta)\sin(\phi) & \cos(\theta) \\ \cos(\theta)\cos(\phi) & \cos(\theta)\sin(\phi) & -\sin(\theta) \end{vmatrix} \\ &= -\sin(\phi)\hat{\mathbf{x}} + \cos(\phi)\hat{\mathbf{y}} \\ &= \hat{\boldsymbol{\phi}}\end{aligned}$$

Finally, the inverse formulas. We may write our existing formulas as follows:

$$\begin{pmatrix} \sin(\theta)\cos(\phi) & \sin(\theta)\sin(\phi) & \cos(\theta) \\ \cos(\theta)\cos(\phi) & \cos(\theta)\sin(\phi) & -\sin(\theta) \\ -\sin(\phi) & \cos(\phi) & 0 \end{pmatrix} \begin{pmatrix} \hat{\mathbf{x}} \\ \hat{\mathbf{y}} \\ \hat{\mathbf{z}} \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{r}} \\ \hat{\boldsymbol{\theta}} \\ \hat{\boldsymbol{\phi}} \end{pmatrix}$$

Now, take note of this insight: the rows of the coefficient matrix form an **orthonormal basis**, rendering the matrix itself **orthogonal**. Therefore it satisfies the property

$$A^{-1} = A^T$$

Ergo, we can immediately write

$$\begin{pmatrix} \hat{\mathbf{x}} \\ \hat{\mathbf{y}} \\ \hat{\mathbf{z}} \end{pmatrix} = \begin{pmatrix} \sin(\theta)\cos(\phi) & \cos(\theta)\cos(\phi) & -\sin(\phi) \\ \sin(\theta)\sin(\phi) & \cos(\theta)\sin(\phi) & \cos(\phi) \\ \cos(\theta) & -\sin(\theta) & 0 \end{pmatrix} \begin{pmatrix} \hat{\mathbf{r}} \\ \hat{\boldsymbol{\theta}} \\ \hat{\boldsymbol{\phi}} \end{pmatrix}$$

which leaves us with

$$\begin{aligned}\hat{\mathbf{x}} &= \sin(\theta)\cos(\phi)\hat{\mathbf{r}} + \cos(\theta)\cos(\phi)\hat{\boldsymbol{\theta}} - \sin(\phi)\hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} &= \sin(\theta)\sin(\phi)\hat{\mathbf{r}} + \cos(\theta)\sin(\phi)\hat{\boldsymbol{\theta}} + \cos(\phi)\hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} &= \cos(\theta)\hat{\mathbf{r}} - \sin(\theta)\hat{\boldsymbol{\theta}}\end{aligned}$$

Problem: 1.39

- Check the divergence theorem for the function $\mathbf{v}_1 = r^2\hat{\mathbf{r}}$, using as your volume the sphere of radius R , centered at the origin.
- Do the same for $\mathbf{v}_2 = (1/r^2)\hat{\mathbf{r}}$. (If the answer surprises you, look back at Prob. 1.16.)

(a) First calculate the integrand

$$\begin{aligned}\nabla \cdot \mathbf{v}_1 &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^4) \\ &= 4r\end{aligned}$$

which we evaluate as

$$\int_V 4r \, d\tau = \int_0^{2\pi} \int_0^\pi \int_0^R 4r^3 \sin(\theta) \, dr \, d\theta \, d\phi = 4\pi R^4$$

For the surface integral, compute

$$\oint_S \mathbf{v}_1 \cdot d\mathbf{a} = \int_0^{2\pi} \int_0^\pi R^4 \sin(\theta) \, d\theta \, d\phi = 4\pi R^4$$

Hence the divergence theorem holds.

(b) The divergence is given by

$$\nabla \cdot \mathbf{v}_2 = \frac{1}{r^2} \frac{\partial}{\partial r} (1) = 0$$

And so the volume integral is correspondingly

$$\int_V 0 \, d\tau = 0$$

Similar to problem 1.16, the evaluation of the divergence is zero, but only because the divergence is undefined at $r = 0$. In truth, the divergence is infinitely concentrated at the origin. In a future section on the three-dimensional delta function, we will see that the divergence is actually $4\pi\delta^3(\mathbf{r})$.

The surface integral evaluates to

$$\oint_S \mathbf{v}_2 \cdot d\mathbf{a} = \int_0^{2\pi} \int_0^\pi \sin(\theta) \, d\theta \, d\phi = 4\pi$$

Problem: 1.40

Compute the divergence of the function

$$\mathbf{v} = r \cos(\theta) \hat{\mathbf{r}} + r \sin(\theta) \hat{\boldsymbol{\theta}} + r \sin(\theta) \cos(\phi) \hat{\boldsymbol{\phi}}$$

Check the divergence theorem for this function, using as your volume the inverted hemispherical bowl of radius R , resting on the xy plane and centered at the origin (Fig. 1.40).

$$\begin{aligned}\nabla \cdot \mathbf{v} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^3 \cos(\theta)) + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \theta} (r \sin^2(\theta)) + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi} (r \sin(\theta) \cos(\phi)) \\ &= 5 \cos(\theta) - \sin(\phi)\end{aligned}$$

The volume integral is given by

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{v} \, d\tau = \int_0^{2\pi} \int_0^{\pi/2} \int_0^R r^2 \sin(\theta) [5 \cos(\theta) - \sin(\phi)] \, dr \, d\theta \, d\phi = \frac{5\pi R^3}{3}$$

and the surface integral

$$\begin{aligned} \oint_S \mathbf{v} \cdot d\mathbf{a} &= \int_0^{2\pi} \int_0^{\pi/2} R^3 \sin(\theta) \cos(\theta) \, d\theta \, d\phi + \int_0^{2\pi} \int_0^R r^2 \, dr \, d\phi \\ &= \pi R^3 + \frac{2\pi R^3}{3} \\ &= \frac{5\pi R^3}{3} \end{aligned}$$

verifying the divergence theorem.

Problem: 1.41

Compute the gradient and Laplacian of the function $T = r[\cos(\theta) + \sin(\theta) \cos(\phi)]$. Check the Laplacian by converting T to Cartesian coordinates and using Eq. 1.42. Test the gradient theorem for this function, using the path shown in Fig. 1.41, from $(0, 0, 0)$ to $(0, 0, 2)$.

$$\begin{aligned} \nabla T &= [\cos(\theta) + \sin(\theta) \cos(\phi)] \hat{\mathbf{r}} + [\cos(\theta) \cos(\phi) - \sin(\theta)] \hat{\boldsymbol{\theta}} - \sin(\phi) \hat{\boldsymbol{\phi}} \\ \nabla^2 T &= \frac{2}{r} [\cos(\theta) + \sin(\theta) \cos(\phi)] + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} (r [\sin(\theta) \cos(\theta) \cos(\phi) - \sin^2(\theta)]) \\ &\quad - \frac{\cos(\phi)}{r \sin(\theta)} \\ &= \frac{2}{r} [\cos(\theta) + \sin(\theta) \cos(\phi)] + \frac{1}{r \sin(\theta)} [(\cos^2(\theta) - \sin^2(\theta)) \cos(\phi) - 2 \sin(\theta) \cos(\theta)] \\ &\quad - \frac{\cos(\phi)}{r \sin(\theta)} \\ &= 0 \end{aligned}$$

In Cartesian coordinates,

$$T = x + z$$

Which we can simply calculate the Laplacian as

$$\nabla^2 T = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0$$

A lesson learned: computations can be made extremely simply or extremely complicated depending on choice of coordinate system. Lastly, by the gradient theorem (path independence), we compute (in Cartesian coordinates)

$$\begin{aligned} \int_{\mathbf{a}}^{\mathbf{b}} (\nabla T) \cdot d\mathbf{l} &= \int_0^2 dz \\ &= T(0, 0, 2) - T(0, 0, 0) \\ &= 2 \end{aligned}$$

1.4.2 - Cylindrical Coordinates

Problem: 1.42

Express the cylindrical unit vectors $\hat{\mathbf{s}}, \hat{\boldsymbol{\phi}}, \hat{\mathbf{z}}$ in terms of $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ (that is, derive Eq. 1.75). “Invert” your formulas to get $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$ in terms of $\hat{\mathbf{s}}, \hat{\boldsymbol{\phi}}, \hat{\mathbf{z}}$ (and ϕ).

As a conceptual matter, note that $\hat{\mathbf{s}}$ and $\hat{\boldsymbol{\phi}}$ live in the xy plane. Using Equation 1.74, we have

$$\hat{\mathbf{s}} = \frac{\mathbf{s}}{|\mathbf{s}|} = \cos(\phi) \hat{\mathbf{x}} + \sin(\phi) \hat{\mathbf{y}}$$

where $|\mathbf{s}| = s$. To find the orthogonal unit vector in ϕ , differentiate $\hat{\mathbf{s}}$ with respect to ϕ and normalize to derive

$$\hat{\boldsymbol{\phi}} = \frac{\partial \hat{\mathbf{s}} / \partial \phi}{|\partial \hat{\mathbf{s}} / \partial \phi|} = -\sin(\phi) \hat{\mathbf{x}} + \cos(\phi) \hat{\mathbf{y}}$$

where $|\partial \hat{\mathbf{s}} / \partial \phi| = 1$. The easiest is $\hat{\mathbf{z}}$:

$$\hat{\mathbf{z}} = \hat{\mathbf{z}}$$

Writing the coordinate transformation in matrix form:

$$\begin{pmatrix} \cos(\phi) & \sin(\phi) & 0 \\ -\sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{\mathbf{x}} \\ \hat{\mathbf{y}} \\ \hat{\mathbf{z}} \end{pmatrix} = \begin{pmatrix} \hat{\mathbf{s}} \\ \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} \end{pmatrix}$$

Once again appealing to orthogonality of the coefficient matrix, we can see that its inverse is equal to its transpose, and thus write

$$\begin{pmatrix} \hat{\mathbf{x}} \\ \hat{\mathbf{y}} \\ \hat{\mathbf{z}} \end{pmatrix} = \begin{pmatrix} \cos(\phi) & -\sin(\phi) & 0 \\ \sin(\phi) & \cos(\phi) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{\mathbf{s}} \\ \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} \end{pmatrix}$$

which produces inverse transformations

$$\begin{aligned} \hat{\mathbf{x}} &= \cos(\phi) \hat{\mathbf{s}} - \sin(\phi) \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{y}} &= \sin(\phi) \hat{\mathbf{s}} + \cos(\phi) \hat{\boldsymbol{\phi}} \\ \hat{\mathbf{z}} &= \hat{\mathbf{z}} \end{aligned}$$

Problem: 1.43

- (a) Find the divergence of the function

$$\mathbf{v} = s(2 + \sin^2(\phi)) \hat{\mathbf{s}} + s \sin(\phi) \cos(\phi) \hat{\phi} + 3z \hat{\mathbf{z}}$$

- (b) Test the divergence theorem for this function, using the quarter-cylinder (radius 2, height 5) shown in Fig. 1.43.
- (c) Find the curl of $\mathbf{v}$.

- (a)

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{1}{s} \frac{\partial}{\partial s} [s^2 (2 + \sin^2(\phi))] + \frac{1}{s} \frac{\partial}{\partial \phi} [s \sin(\phi) \cos(\phi)] + \frac{\partial}{\partial z} (3z) \\ &= 4 + 2 \sin^2(\phi) + \cos^2(\phi) - \sin^2(\phi) + 3 \\ &= 8 \end{aligned}$$

- (b) The volume integral is

$$\int_V \nabla \cdot \mathbf{v} \, d\tau = \int_0^5 \int_0^{\pi/2} \int_0^2 8s \, ds \, d\phi \, dz = 40\pi$$

and the surface integral, after formulating infinitesimal areas $s \, ds \, d\phi \, \hat{\mathbf{z}}$, $ds \, dz \, \hat{\phi}$, and $s \, d\phi \, dz \, \hat{\mathbf{s}}$ (where the variable held constant corresponds to the unit vector), is given by

$$\begin{aligned} \oint_S \mathbf{v} \cdot d\mathbf{a} &= \int_0^{\pi/2} \int_0^2 3 \cdot 5s \, ds \, d\phi + \int_0^5 \int_0^{\pi/2} 2^2 (2 + \sin^2(\phi)) \, d\phi \, dz \\ &= 15\pi + 25\pi \\ &= 40\pi \end{aligned}$$

verifying that the divergence theorem works.

- (c)

$$\nabla \times \mathbf{v} = \frac{1}{s} \left[\frac{\partial}{\partial s} (s^2 \sin(\phi) \cos(\phi)) - \frac{\partial}{\partial \phi} s(2 + \sin^2(\phi)) \right] \hat{\mathbf{z}} = \mathbf{0}$$

1.5 - The Dirac Delta Function

1.5.2 - The One-Dimensional Dirac Delta Function

Problem: 1.44

Evaluate the following integrals:

(a) $\int_2^6 (3x^2 - 2x - 1) \delta(x - 3) dx$

(b) $\int_0^5 \cos(x) \delta(x - \pi) dx$

(c) $\int_0^3 x^3 \delta(x + 1) dx$

(d) $\int_{-\infty}^{\infty} \ln(x + 3) \delta(x + 2) dx$

(a) 20

(b) -1

(c) 0

(d) 0

Problem: 1.45

Evaluate the following integrals:

(a) $\int_{-2}^2 (2x + 3) \delta(3x) dx$

(b) $\int_0^2 (x^3 + 3x + 2) \delta(1 - x) dx$

(c) $\int_{-1}^1 9x^2 \delta(3x + 1) dx$

(d) $\int_{-\infty}^a \delta(x - b) dx$

(a) 1

(b) 6

(c) $\frac{1}{3}$ ($\delta(3x + 1) = 0$ at $x = -1/3$ and the $1/3$ scaling factor comes out)

(d)

$$\int_{-\infty}^a \delta(x - b) dx = \begin{cases} 1 & b \in (-\infty, a] \\ 0 & b \notin (-\infty, a] \end{cases}$$

Problem: 1.46

(a) Show that

$$x \frac{d}{dx}(\delta(x)) = -\delta(x)$$

[Hint: Use integration by parts.]

(b) Let $\theta(x)$ be the **step function**:

$$\theta(x) \equiv \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$$

Show that $d\theta/dx = \delta(x)$.

(a) Let $u = x$ and $dv = \frac{d}{dx}(\delta(x)) dx$. Then $du = dx$ and $v = \delta(x)$, which implies

$$\int_{-\infty}^{\infty} x \frac{d}{dx}(\delta(x)) dx = \underbrace{x\delta(x)}_{=0} \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta(x) dx = - \int_{-\infty}^{\infty} \delta(x) dx$$

Equating the integrands produces the desired result.

(b) The intuition here is to proceed with integration by parts. Consider the function $f(x)$. Compute, with $u = f(x)$ and $dv = (d\theta/dx) dx$:

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) \frac{d\theta}{dx} dx &= f(x) \theta(x) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \theta(x) \frac{df}{dx} dx \\ &= f(\infty) - \int_0^{\infty} \frac{df}{dx} dx \\ &= f(0) \end{aligned}$$

where the second equality arises from the definition of $\theta(x)$, and the third from the Fundamental Theorem of Calculus. Now, the equality is satisfied if and only if

$$\frac{d\theta}{dx} = \delta(x)$$

1.5.3 - The Three-Dimensional Delta Function

Problem: 1.47

- (a) Write an expression for the volume charge density $\rho(\mathbf{r})$ of a point charge q at $\mathbf{r}'$. Make sure that the volume integral of ρ equals q .
- (b) What is the volume charge density of an electric dipole, consisting of a point charge $-q$ at the origin and a point charge $+q$ at $\mathbf{a}$?
- (c) What is the volume charge density (in spherical coordinates) of a uniform, infinitesimally thin spherical shell of radius R and total charge Q , centered at the origin? [*Beware*: the integral over all space must equal Q .]

- (a) One approach is to take inspiration from the vector function

$$\mathbf{v} = \frac{1}{r^2} \hat{\mathbf{r}}$$

and replacing $\hat{\mathbf{r}}$ or r with $\hat{\mathbf{z}}$ or z , multiplying by $q/4\pi$, and taking the divergence, concluding

$$\rho(\mathbf{z}) = \nabla \cdot \frac{q}{4\pi z^2} \hat{\mathbf{z}} = q\delta^3(\mathbf{z})$$

then we can see that

$$\int_V \nabla \cdot \frac{q}{4\pi z^2} \hat{\mathbf{z}} d\tau = \int_V q\delta^3(\mathbf{z}) d\tau = q$$

- (b) By superposition,

$$\rho(\mathbf{r}) = \nabla \cdot \frac{q}{4\pi |\mathbf{r} - \mathbf{a}|^3} (\mathbf{r} - \mathbf{a}) - \nabla \cdot \frac{q}{4\pi r^2} \hat{\mathbf{r}} = q[\delta^3(\mathbf{r} - \mathbf{a}) - \delta^3(\mathbf{r})]$$

where the cubed term in the denominator of the first term is necessary to normalize the accompanying vector.

- (c) The charge, as opposed to being concentrated at a point, is distributed across the shell. One approach is to work backwards from the constraint

$$\int_V \rho(r, \theta, \phi) d\tau = \int_0^{2\pi} \int_0^\pi \int_0^\infty \rho(r, \theta, \phi) r^2 \sin(\theta) dr d\theta d\phi = Q$$

If we require $\rho(r, \theta, \phi) = \rho(r)$ (strictly a function of r) and simplify the volume integral slightly, we have

$$\int_0^\infty 4\pi \rho(r) r^2 dr = Q$$

which is solved by

$$\rho(r) = \frac{Q}{4\pi R^2} \delta(r - R)$$

Problem: 1.48

Evaluate the following integrals:

- (a) $\int (r^2 + \mathbf{r} \cdot \mathbf{a} + a^2) \delta^3(\mathbf{r} - \mathbf{a}) d\tau$, where $\mathbf{a}$ is a fixed vector, a is its magnitude, and the integral is over all space.
- (b) $\int_{\mathcal{V}} |\mathbf{r} - \mathbf{b}|^2 \delta^3(5\mathbf{r}) d\tau$, where $\mathcal{V}$ is a cube of side 2, centered on the origin, and $\mathbf{b} = 4\hat{\mathbf{y}} + 3\hat{\mathbf{z}}$.
- (c) $\int_{\mathcal{V}} [r^4 + r^2(\mathbf{r} \cdot \mathbf{c}) + c^4] \delta^3(\mathbf{r} - \mathbf{c}) d\tau$, where $\mathcal{V}$ is a sphere of radius 6 about the origin, $\mathbf{c} = 5\hat{\mathbf{x}} + 3\hat{\mathbf{y}} + 2\hat{\mathbf{z}}$, and c is its magnitude.
- (d) $\int_{\mathcal{V}} \mathbf{r} \cdot (\mathbf{d} - \mathbf{r}) \delta^3(\mathbf{e} - \mathbf{r}) d\tau$, where $\mathbf{d} = (1, 2, 3)$, $\mathbf{e} = (3, 2, 1)$, and $\mathcal{V}$ is a sphere of radius 1.5 centered at $(2, 2, 2)$.

(a) $3a^2$

(b) $1/5$, using the scaling factor

$$\delta^3(k\mathbf{r}) = \frac{1}{|k|^3} \delta^3(\mathbf{r})$$

(c) Since $|\mathbf{c}| = \sqrt{38} > 6$, the spike falls outside of the sphere, and the volume integral vanishes.

(d) Since $|\mathbf{o} - \mathbf{e}| = \sqrt{2} < 1.5$ (where $\mathbf{o}$ is the center of the sphere), $\mathbf{e}$ falls within the volume defined by $\mathcal{V}$, and the volume integral evaluates to

$$\mathbf{e} \cdot (\mathbf{d} - \mathbf{e}) = 3(-2) + 2(0) + 1(2) = -4$$

Problem: 1.49

Evaluate the integral

$$J = \int_{\mathcal{V}} e^{-r} \left(\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau$$

(where $\mathcal{V}$ is a sphere of radius R , centered at the origin) by two different methods, as in Ex. 1.16.

Method 1:

$$J = \int_{\mathcal{V}} e^{-r} \left(\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau = \int_{\mathcal{V}} e^{-r} 4\pi \delta^3(\mathbf{r}) d\tau = 4\pi$$

Method 2:

$$J = \int_{\mathcal{V}} e^{-r} \left(\nabla \cdot \frac{\hat{\mathbf{r}}}{r^2} \right) d\tau = - \int_{\mathcal{V}} \frac{\hat{\mathbf{r}}}{r^2} \cdot (\nabla e^{-r}) d\tau + \oint_{\mathcal{S}} e^{-r} \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{a}$$

First, evaluate the volume integral:

$$\begin{aligned} - \int_V \frac{\hat{\mathbf{r}}}{r^2} \cdot (\nabla e^{-r}) \, d\tau &= \int_0^{2\pi} \int_0^\pi \int_0^R e^{-r} \sin(\theta) \, dr \, d\theta \, d\phi \\ &= 4\pi(1 - e^{-R}) \end{aligned}$$

And then the surface integral:

$$\oint_S e^{-r} \frac{\hat{\mathbf{r}}}{r^2} \cdot d\mathbf{a} = \int_0^{2\pi} \int_0^\pi e^{-R} \sin(\theta) \, d\theta \, d\phi = 4\pi e^{-R}$$

Summing produces 4π .

1.6 - The Theory of Vector Fields

1.6.2 - Potentials

Problem: 1.50

- (a) Let $\mathbf{F}_1 = x^2\hat{\mathbf{z}}$ and $\mathbf{F}_2 = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$. Calculate the divergence and curl of $\mathbf{F}_1$ and $\mathbf{F}_2$. Which one can be written as the gradient of a scalar? Find a scalar potential that does the job. Which one can be written as the curl of a vector? Find a suitable vector potential.
- (b) Show that $\mathbf{F}_3 = yz\hat{\mathbf{x}} + zx\hat{\mathbf{y}} + xy\hat{\mathbf{z}}$ can be written as both the gradient of a scalar and as the curl of a vector. Find scalar and vector potentials for this function.

(a)

$$\begin{aligned}\nabla \cdot \mathbf{F}_1 &= \frac{\partial}{\partial z}(x^2) = 0 \\ \nabla \times \mathbf{F}_1 &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & x^2 \end{vmatrix} \\ &= -2x\hat{\mathbf{y}} \\ \nabla \cdot \mathbf{F}_2 &= \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 3 \\ \nabla \times \mathbf{F}_2 &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} \\ &= \mathbf{0}\end{aligned}$$

$\mathbf{F}_2$ can be written as the gradient of a scalar and $\mathbf{F}_1$ the curl of a vector. One scalar potential for $\mathbf{F}_2$ is

$$V = -\frac{1}{2}(x^2 + y^2 + z^2)$$

which produces

$$\mathbf{F}_2 = -\nabla V$$

and a vector potential for $\mathbf{F}_1$:

$$\mathbf{A} = \frac{x^3}{3}\hat{\mathbf{z}}$$

which produces

$$\mathbf{F}_1 = \nabla \times \mathbf{A}$$

(b)

$$\begin{aligned}\nabla \cdot \mathbf{F}_3 &= \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(zx) + \frac{\partial}{\partial z}(xy) = 0 \\ \nabla \times \mathbf{F}_3 &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & zx & xy \end{vmatrix} \\ &= \mathbf{0}\end{aligned}$$

Scalar potential:

$$V = -xyz$$

Vector potential: The vector potential $\mathbf{A}$ must satisfy these properties:

$$\begin{aligned}\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} &= yz \\ \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} &= zx \\ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} &= xy\end{aligned}$$

To generate $\mathbf{A}$, for the sake of simplicity, set $A_z = 0$. Then one proposal satisfying the first two equations is

$$A_x = \frac{xz^2}{2}, \quad A_y = -\frac{yz^2}{2}$$

The last equation, however, is unsatisfied. To correct our solution, we can add a $-xy^2/2$ term to A_x , which is not a function of z so the second equation remains satisfied. Now all three equations are satisfied, and we have

$$\mathbf{A} = \frac{x}{2}(z^2 - y^2)\hat{\mathbf{x}} - \frac{1}{2}yz^2\hat{\mathbf{y}}$$

Problem: 1.51

For Theorem 1, show that (d) $\implies$ (a), (a) $\implies$ (c), (c) $\implies$ (b), (b) $\implies$ (c), and (c) $\implies$ (a).

PROOF. (d) $\implies$ (a): $\nabla \times \mathbf{F} = \nabla \times (-\nabla V) = \mathbf{0}$ since the curl of a gradient always vanishes.

(a) $\implies$ (c): By Stokes' theorem,

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{a} = \oint_{\mathcal{P}} \mathbf{F} \cdot d\mathbf{l} = 0$$

if $\nabla \times \mathbf{F} = \mathbf{0}$.

(c) $\implies$ (b): Take any two arbitrary points; call them $\mathbf{a}$ and $\mathbf{b}$. Consider two paths between them: $\mathcal{P}_1$ and $\mathcal{P}_2$. Grafting the ends together, they form a closed loop:

$$\oint = \int_{\mathcal{P}_1} - \int_{\mathcal{P}_2} = 0$$

where the negative sign arises from the need to keep consistent our direction of movement along the loop. By assumption, the difference is zero, thus

$$\int_{\mathcal{P}_1} = \int_{\mathcal{P}_2}$$

which is akin to saying $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l}$ is independent of path.

(b) $\implies$ (c): Let $\mathbf{a} = \mathbf{b}$. Then by path independence, we can take the difference of the integral at equal endpoints, and see that the closed loop integral vanishes.

$$\int_{\mathbf{a}}^{\mathbf{a}} \mathbf{F} \cdot d\mathbf{l} = \oint_P \mathbf{F} \cdot d\mathbf{l} = 0$$

(c) $\implies$ (a): By Stokes' theorem,

$$\oint \mathbf{F} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{F}) \cdot d\mathbf{a} = 0$$

Physically, the surface integral measures the flux of the curl through the surface. If there is no flux given that we have zero for any arbitrary loop, then it must follow that there is no curl:

$$\nabla \times \mathbf{F} = \mathbf{0}$$

□

Problem: 1.52

For Theorem 2, show that (d) $\implies$ (a), (a) $\implies$ (c), (c) $\implies$ (b), (b) $\implies$ (c), and (c) $\implies$ (a).

PROOF. (d) $\implies$ (a): $\nabla \cdot \mathbf{F} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$ as the divergence of a curl is always zero.

(a) $\implies$ (c): By the divergence theorem,

$$\int_V (\nabla \cdot \mathbf{F}) \cdot d\tau = \oint_S \mathbf{F} \cdot d\mathbf{a} = 0$$

(c) $\implies$ (b): This is the three-dimensional analogue to Theorem 1. Consider any perimeter around $\mathcal{S}$. Slice through it to expose an opening in two surfaces $\mathcal{S}_1$ and $\mathcal{S}_2$. Then we can write

$$\oint_{\mathcal{S}} = \int_{\mathcal{S}_1} - \int_{\mathcal{S}_2} = 0 \implies \int_{\mathcal{S}_1} = \int_{\mathcal{S}_2}$$

Establishing independence of $\int \mathbf{F} \cdot d\mathbf{a}$ of surface.

(b) $\implies$ (c): Let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two open surfaces that share the same boundary. Melding them at the boundary produces a closed surface. Summing the fluxes from each produces

$$\int_{\mathcal{S}_1} - \int_{\mathcal{S}_2} = \oint_{\mathcal{S}} = 0$$

where the zero equality emerges from the independence of surface.

(c) $\implies$ (a): By the divergence theorem,

$$\oint \mathbf{F} \cdot d\mathbf{a} = \int_V (\nabla \cdot \mathbf{F}) d\tau = 0$$

Since the volume integral of the divergence is zero for any arbitrary surface, it must be that $\nabla \cdot \mathbf{F} = 0$. $\square$

Problem: 1.53

- (a) Which of the vectors in Problem 1.15 can be expressed as the gradient of a scalar? Find a scalar function that does the job.
- (b) Which can be expressed as the curl of a vector? Find such a vector.

(a) $\nabla \times \mathbf{v}_c = \mathbf{0}$, $\mathbf{v}_c = -\nabla V$ where $V = -xy^2 - yz^2$

(b) $\nabla \cdot \mathbf{v}_a = 0$, $\mathbf{v}_a = \nabla \times \mathbf{A}$ where $\mathbf{A} = xz^3\hat{\mathbf{x}} - x^2z\hat{\mathbf{y}}$

More Problems on Chapter 1

Problem: 1.54

Check the divergence theorem for the function

$$\mathbf{v} = r^2 \cos(\theta) \hat{\mathbf{r}} + r^2 \cos(\phi) \hat{\boldsymbol{\theta}} - r^2 \cos(\theta) \sin(\phi) \hat{\boldsymbol{\phi}}$$

using as your volume one octant of the sphere of radius R (Fig. 1.48). Make sure you include the *entire* surface.

First calculating the divergence:

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{1}{r^2} \frac{\partial}{\partial r} [r^4 \cos(\theta)] + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \theta} [r^2 \sin(\theta) \cos(\phi)] + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi} [-r^2 \cos(\theta) \sin(\phi)] \\ &= 4r \cos(\theta) + r \cot(\theta) \cos(\phi) - r \cot(\theta) \cos(\phi) \\ &= 4r \cos(\theta) \end{aligned}$$

The volume integral is

$$\begin{aligned} \int_V 4r \cos(\theta) d\tau &= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^R 4r^3 \sin(\theta) \cos(\theta) dr d\theta d\phi \\ &= \frac{\pi R^4}{4} \end{aligned}$$

And the surface integral

$$\begin{aligned} \oint_S \mathbf{v} \cdot d\mathbf{a} &= \underbrace{\int_0^{\pi/2} \int_0^{\pi/2} R^4 \sin(\theta) \cos(\theta) d\theta d\phi}_{\text{curved surface}} + \underbrace{\int_0^{\pi/2} \int_0^R r^3 \cos(\phi) dr d\phi}_{\text{xy surface}} \\ &\quad - \underbrace{\int_0^{\pi/2} \int_0^R r^3 \cos(\theta) \sin(0) d\theta dr}_{\text{xz surface, } \phi=0} - \underbrace{\int_0^{\pi/2} \int_0^R r^3 \cos(\theta) \sin\left(\frac{\pi}{2}\right) dr d\theta}_{\text{yz surface, } \phi=\pi/2} \\ &= \frac{\pi R^4}{4} + \frac{R^4}{4} - 0 - \frac{R^4}{4} \\ &= \frac{\pi R^4}{4} \end{aligned}$$

Problem: 1.55

Check Stokes' theorem using the function $\mathbf{v} = ay\hat{\mathbf{x}} + bx\hat{\mathbf{y}}$ (a and b are constants) and the circular path of radius R , centered at the origin in the xy plane.

First convert to cylindrical coordinates:

$$\begin{aligned} \mathbf{v} &= s \left(a \sin(\phi) [\cos(\phi) \hat{\mathbf{s}} - \sin(\phi) \hat{\boldsymbol{\phi}}] + b \cos(\phi) [\sin(\phi) \hat{\mathbf{s}} + \cos(\phi) \hat{\boldsymbol{\phi}}] \right) \\ &= s \sin(\phi) \cos(\phi) (a + b) \hat{\mathbf{s}} + s [b \cos^2(\phi) - a \sin^2(\phi)] \hat{\boldsymbol{\phi}} \\ &= \frac{s(a+b)}{2} \sin(2\phi) \hat{\mathbf{s}} + s \left[\left(\frac{b-a}{2} \right) + \left(\frac{a+b}{2} \right) \cos(2\phi) \right] \hat{\boldsymbol{\phi}} \end{aligned}$$

And calculate the curl:

$$\begin{aligned}\nabla \times \mathbf{v} &= \frac{1}{s} \left[\frac{\partial}{\partial s} \left(s^2 \left[\left(\frac{b-a}{2} \right) + \left(\frac{a+b}{2} \right) \cos(2\phi) \right] \right) - \frac{\partial}{\partial \phi} \left(\frac{s(a+b)}{2} \sin(2\phi) \right) \right] \hat{\mathbf{z}} \\ &= [(b-a) + (a+b) \cos(2\phi) - (a+b) \cos(2\phi)] \hat{\mathbf{z}} \\ &= (b-a) \hat{\mathbf{z}}\end{aligned}$$

In calculating the surface integral, no surface is specified in the problem. Therefore, since from Stokes' theorem the surface integral is dependent only on the boundary line, we can pick the simplest surface: the circular surface with radius R on the xy plane. Compute

$$\int_S (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^{2\pi} \int_0^R (b-a) s \, ds \, d\phi = (b-a) \pi R^2$$

Now, the line integral

$$\begin{aligned}\oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l} &= \int_0^R \frac{s(a+b)}{2} \sin(4\pi) \, ds + \int_0^{2\pi} R^2 \left[\left(\frac{b-a}{2} \right) + \left(\frac{a+b}{2} \right) \cos(2\phi) \right] d\phi \\ &= (b-a) \pi R^2\end{aligned}$$

Problem: 1.56

Compute the line integral of

$$\mathbf{v} = 6x\hat{\mathbf{x}} + yz^2\hat{\mathbf{y}} + (3y + z)\hat{\mathbf{z}}$$

along the triangular path shown in Fig. 1.49. Check your answer using Stokes' theorem.

$$(0, 0, 0) \rightarrow (0, 1, 0): x = z = 0, d\mathbf{l} = dy\hat{\mathbf{y}}, \mathbf{v} \cdot d\mathbf{l} = 0$$

$$\int_{\mathcal{P}_1} \mathbf{v} \cdot d\mathbf{l} = 0$$

$$(0, 1, 0) \rightarrow (0, 0, 2):$$

$$x = 0$$

$$z = -2y + 2$$

$$d\mathbf{l} = dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}} = dy(\hat{\mathbf{y}} - 2\hat{\mathbf{z}})$$

$$\begin{aligned}\mathbf{v} \cdot d\mathbf{l} &= (yz^2 - 6y - 2z) \, dy \\ &= (4y^3 - 8y^2 + 2y - 4) \, dy\end{aligned}$$

$$\int_{\mathcal{P}_2} \mathbf{v} \cdot d\mathbf{l} = \int_1^0 (4y^3 - 8y^2 + 2y - 4) \, dy = -1 + \frac{8}{3} - 1 + 4 = \frac{14}{3}$$

$$(0, 0, 2) \rightarrow (0, 0, 0): x = y = 0, d\mathbf{l} = dz\hat{\mathbf{z}}$$

$$\int_{\mathcal{P}_3} \mathbf{v} \cdot d\mathbf{l} = \int_2^0 z \, dz = -2$$

Summing produces

$$\oint_{\mathcal{P}} \mathbf{v} \cdot d\mathbf{l} = \frac{8}{3}$$

Using Stokes' theorem, consider the triangular surface spanned in Figure 1.49. We have infinitesimal area $d\mathbf{a} = dz dy \hat{\mathbf{x}}$ with bounds $z \in [0, -2y + 2]$ and $y \in [0, 1]$. The curl is

$$\nabla \times \mathbf{v} = (3 - 2yz) \hat{\mathbf{x}}$$

Compute

$$\int_0^1 \int_0^{-2y+2} (3 - 2yz) dz dy = \int_1^0 (-4y^3 + 8y^2 - 10y + 6) dy = -1 + \frac{8}{3} - 5 + 6 = \frac{8}{3}$$

Problem: 1.57

Compute the line integral of

$$\mathbf{v} = r \cos^2(\theta) \hat{\mathbf{r}} - r \cos(\theta) \sin(\theta) \hat{\boldsymbol{\theta}} + 3r \hat{\boldsymbol{\phi}}$$

around the path shown in Fig. 1.50 (the points are labeled by their Cartesian coordinates). Do it either in cylindrical or in spherical coordinates. Check your answer, using Stokes' theorem.

There is a great deal of tedium in evaluating each segment separately. One slick approach is to observe that in cylindrical coordinates, we have (evaluated a few lines below)

$$\mathbf{v} = z \hat{\mathbf{z}} + 3\sqrt{s^2 + z^2} \hat{\boldsymbol{\phi}}$$

By the fundamental theorem of gradients, we can observe that the $\hat{\mathbf{z}}$ component is in fact expressible as a gradient, implying that the path contribution from that component must be zero all around the loop:

$$z \hat{\mathbf{z}} = \nabla \left(\frac{z^2}{2} \right)$$

We will see this hold true in the tedium of evaluating the line integral for each component. In general, the infinitesimal displacement in spherical coordinates is

$$d\mathbf{l} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin(\theta) d\phi \hat{\boldsymbol{\phi}}$$

and in cylindrical

$$d\mathbf{l} = ds \hat{\mathbf{s}} + s d\phi \hat{\boldsymbol{\phi}} + dz \hat{\mathbf{z}}$$

Furthermore, $\mathbf{v}$ in cylindrical coordinates is

$$\begin{aligned} \mathbf{v} &= r \cos(\theta) [\cos(\theta) \hat{\mathbf{r}} - \sin(\theta) \hat{\boldsymbol{\theta}}] + 3r \hat{\boldsymbol{\phi}} \\ &= r \cos(\theta) \hat{\mathbf{z}} + 3r \hat{\boldsymbol{\phi}} \\ &= z \hat{\mathbf{z}} + 3\sqrt{s^2 + z^2} \hat{\boldsymbol{\phi}} \end{aligned}$$

The strategy is to identify which basis is ideal for each segment: spherical or cylindrical. Starting at the origin, calculate each segment's contribution:

Spherical: $(0, 0, 0) \rightarrow (1, 0, 0): \theta = \pi/2, \phi = 0, d\mathbf{l} = dr \hat{\mathbf{r}}, \mathbf{v} \cdot d\mathbf{l} = r \cos^2(\pi/2) dr = 0$

$$\int_{\mathcal{P}_1} \mathbf{v} \cdot d\mathbf{l} = 0$$

Spherical: $(1, 0, 0) \rightarrow (0, 1, 0): \theta = \pi/2, r = 1, d\mathbf{l} = r \sin(\pi/2) d\phi \hat{\boldsymbol{\phi}} = d\phi \hat{\boldsymbol{\phi}}, \mathbf{v} \cdot d\mathbf{l} = 3 d\phi$

$$\int_{\mathcal{P}_2} \mathbf{v} \cdot d\mathbf{l} = \int_0^{\pi/2} 3 d\phi = \frac{3\pi}{2}$$

Cylindrical: $(0, 1, 0) \rightarrow (0, 1, 2): s = 1, \phi = \pi/2, d\mathbf{l} = dz \hat{\mathbf{z}}, \mathbf{v} \cdot d\mathbf{l} = z dz$

$$\int_{\mathcal{P}_3} \mathbf{v} \cdot d\mathbf{l} = \int_0^2 z dz = 2$$

Cylindrical: $(0, 1, 2) \rightarrow (0, 0, 0): z = 2s, \phi = \pi/2, d\mathbf{l} = ds \hat{\mathbf{s}} + dz \hat{\mathbf{z}} = dz(\frac{1}{2}\hat{\mathbf{s}} + \hat{\mathbf{z}}), \mathbf{v} \cdot d\mathbf{l} = z dz$

$$\int_{\mathcal{P}_4} \mathbf{v} \cdot d\mathbf{l} = \int_2^0 z dz = -2$$

Summing each component produces $3\pi/2$. To summarize, the strategy here is to identify the coordinate system for each segment in which calculations are easiest.

The two surfaces are the quarter circle and the triangle. For the quarter circle,

$$\nabla \times \mathbf{v} = \underbrace{3 \cot(\theta) \hat{\mathbf{r}} - 6\hat{\boldsymbol{\theta}}}_{\text{spherical}} = -\underbrace{\frac{3z}{\sqrt{s^2 + z^2}} \hat{\mathbf{s}} + \frac{3(2s^2 + z^2)}{s\sqrt{s^2 + z^2}} \hat{\mathbf{z}}}_{\text{cylindrical}}$$

Now, the quarter circle lies in the plane where $\theta = \pi/2$ (spherical) or $z = 0$ (cylindrical), so our curls reduce to

$$\nabla \times \mathbf{v} = \underbrace{-6\hat{\boldsymbol{\theta}}}_{\text{spherical}} = \underbrace{6\hat{\mathbf{z}}}_{\text{cylindrical}}$$

The infinitesimal areas are $d\mathbf{a} = -r dr d\phi \hat{\boldsymbol{\theta}}$ (spherical) and $d\mathbf{a} = s ds d\phi \hat{\mathbf{z}}$ (cylindrical). Note that the negative sign in the spherical infinitesimal area arises from the fact that $\hat{\boldsymbol{\theta}}$ always points downwards; at $\theta = \pi/2$ it is antiparallel to $\hat{\mathbf{z}}$. Calculating the surface integral contribution from the quarter circle produces

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = - \int_0^{\pi/2} \int_0^1 (-1) 6r dr d\phi = \int_0^{\pi/2} \int_0^1 6s ds d\phi = \frac{3\pi}{2}$$

In the case of the triangle, the infinitesimal areas point in the $\hat{\boldsymbol{\phi}}$ (spherical and cylindrical) direction, which are orthogonal to the respective curls. Therefore the triangle contribution is zero, and Stokes' theorem is verified.

Problem: 1.58

Check Stokes' theorem for the function $\mathbf{v} = y\hat{\mathbf{z}}$, using the triangular surface shown in Fig. 1.51.

Begin by calculating the line integral.

$$\begin{aligned} (0, 0, a) \rightarrow (a, 0, 0): y = dy = 0, z = -x + a, d\mathbf{l} = dx\hat{\mathbf{x}} + dz\hat{\mathbf{z}} = dx(\hat{\mathbf{x}} - \hat{\mathbf{z}}), \\ \mathbf{v} \cdot d\mathbf{l} = -y dx = 0 \end{aligned}$$

$$\int_{\mathcal{P}_1} \mathbf{v} \cdot d\mathbf{l} = 0$$

$$\begin{aligned} (a, 0, 0) \rightarrow (0, 2a, 0): z = dz = 0, y = 2(a - x), d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} = dx(\hat{\mathbf{x}} - 2\hat{\mathbf{y}}), \\ \mathbf{v} \cdot d\mathbf{l} = 0 \end{aligned}$$

$$\int_{\mathcal{P}_2} \mathbf{v} \cdot d\mathbf{l} = 0$$

$$\begin{aligned} (0, 2a, 0) \rightarrow (0, 0, a): x = dx = 0, y = 2(a - z), d\mathbf{l} = dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}} = dy(\hat{\mathbf{y}} - \tfrac{1}{2}\hat{\mathbf{z}}), \\ \mathbf{v} \cdot d\mathbf{l} = -\tfrac{1}{2}y dy \end{aligned}$$

$$\int_{\mathcal{P}_3} \mathbf{v} \cdot d\mathbf{l} = -\int_{2a}^0 \frac{y}{2} dy = a^2$$

The curl is

$$\nabla \times \mathbf{v} = \hat{\mathbf{x}}$$

Projecting the “shadow” of the triangular surface onto the yz plane (the only non-zero component arising from the dot product of $d\mathbf{a}$ with the curl), we have infinitesimal area $d\mathbf{a} = dy dz \hat{\mathbf{x}}$, and we compute

$$\int (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = \int_0^a \int_0^{2(a-z)} dy dz = \int_0^a 2(a-z) dz = 2a^2 - a^2 = a^2$$

verifying Stokes' theorem.

Problem: 1.59

Check the divergence theorem for the function

$$\mathbf{v} = r^2 \sin(\theta) \hat{\mathbf{r}} + 4r^2 \cos(\theta) \hat{\boldsymbol{\theta}} + r^2 \tan(\theta) \hat{\boldsymbol{\phi}}$$

using the volume of the “ice-cream cone” shown in Fig. 1.52 (the top surface is spherical, with radius R and centered at the origin).

The divergence is given by

$$\begin{aligned} \nabla \cdot \mathbf{v} &= \frac{1}{r^2} \frac{\partial}{\partial r} [r^4 \sin(\theta)] + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \theta} [4r^2 \sin(\theta) \cos(\theta)] + \frac{1}{r \sin(\theta)} \frac{\partial}{\partial \phi} [r^2 \tan(\theta)] \\ &= 4r \sin(\theta) + \frac{4r [\cos^2(\theta) - \sin^2(\theta)]}{\sin(\theta)} \\ &= \frac{4r [\sin^2(\theta) + \cos^2(\theta) - \sin^2(\theta)]}{\sin(\theta)} \\ &= \frac{4r \cos^2(\theta)}{\sin(\theta)} \end{aligned}$$

The volume integral is computed as

$$\begin{aligned}
 \int \nabla \cdot \mathbf{v} \, d\tau &= \int_0^{2\pi} \int_0^{\pi/6} \int_0^R 4r^3 \cos^2(\theta) \, dr \, d\theta \, d\phi \\
 &= 2\pi R^4 \int_0^{\pi/6} \cos^2(\theta) \, d\theta \\
 &= 2\pi R^4 \left[\frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right]_0^{\pi/6} \\
 &= 2\pi R^4 \left(\frac{\pi}{12} + \frac{\sqrt{3}}{8} \right) \\
 &= \frac{\pi R^4}{12} (2\pi + 3\sqrt{3})
 \end{aligned}$$

And the surface integral

$$\begin{aligned}
 \int \mathbf{v} \cdot d\mathbf{a} &= \int_0^{2\pi} \int_0^{\pi/6} R^4 \sin^2(\theta) \, d\theta \, d\phi + \int_0^{2\pi} \int_0^R 4r^3 \cos\left(\frac{\pi}{6}\right) \sin\left(\frac{\pi}{6}\right) \, dr \, d\phi \\
 &= 2\pi R^4 \int_0^{\pi/6} \sin^2(\theta) \, d\theta + 2\sqrt{3}\pi \int_0^R r^3 \, dr \\
 &= 2\pi R^4 \left[\frac{\theta}{2} - \frac{\sin(2\theta)}{4} \right]_0^{\pi/6} + \frac{\sqrt{3}\pi R^4}{2} \\
 &= \frac{\pi R^4}{12} (2\pi - 3\sqrt{3}) + \frac{\sqrt{3}\pi R^4}{2} \\
 &= \frac{\pi R^4}{12} (2\pi + 3\sqrt{3})
 \end{aligned}$$

Confirming the divergence theorem.

Problem: 1.60

Here are two cute checks of the fundamental theorems:

- (a) Combine Corollary 2 to the gradient theorem with Stokes' theorem ($\mathbf{v} = \nabla T$, in this case). Show that the result is consistent with what you already knew about second derivatives.
 - (b) Combine Corollary 2 to Stokes' theorem with the divergence theorem. Show that the result is consistent with what you already knew.
-
- (a) Corollary 2 states that $\oint \nabla T \cdot d\mathbf{l} = 0$. By Stokes' theorem, $\int (\nabla \times \nabla T) \cdot d\mathbf{a} = 0$, which is consistent with the fact that the curl of a gradient is always zero.
 - (b) Corollary 2 states that $\oint (\nabla \times \mathbf{v}) \cdot d\mathbf{a} = 0$. By the divergence theorem, $\int (\nabla \cdot (\nabla \times \mathbf{v})) \, d\tau = 0$, which is consistent with the fact that the divergence of a curl is always zero.

Suppose we had not known these facts. In both cases, it is the arbitrariness of the surface and volume, respectively, that forces the integrand to zero, and establishes the second derivative vanishing identities.

Problem: 1.61

Although the gradient, divergence, and curl theorems are the fundamental integral theorems of vector calculus, it is possible to derive a number of corollaries from them. Show that:

- (a) $\int_V (\nabla T) \, d\tau = \oint_S T \, d\mathbf{a}$. [*Hint:* Let $\mathbf{v} = \mathbf{c}T$, where $\mathbf{c}$ is a constant, in the divergence theorem; use the product rules.]
- (b) $\int_V (\nabla \times \mathbf{v}) \, d\tau = -\oint_S \mathbf{v} \times d\mathbf{a}$. [*Hint:* Replace $\mathbf{v}$ by $(\mathbf{v} \times \mathbf{c})$ in the divergence theorem.]
- (c) $\int_V [T\nabla^2 U + (\nabla T) \cdot (\nabla U)] \, d\tau = \oint_S (T\nabla U) \cdot d\mathbf{a}$. [*Hint:* Let $\mathbf{v} = T\nabla U$ in the divergence theorem.]
- (d) $\int_V (T\nabla^2 U - U\nabla^2 T) \, d\tau = \oint_S (T\nabla U - U\nabla T) \cdot d\mathbf{a}$. [*Comment:* This is sometimes called **Green's second identity**; it follows from (c), which is known as **Green's identity**.]
- (e) $\int_S \nabla T \times d\mathbf{a} = -\oint_P T \, d\mathbf{l}$. [*Hint:* Let $\mathbf{v} = \mathbf{c}T$ in Stokes' theorem.]

- (a) To clarify, $\mathbf{c}$ is a constant *vector*. Let $\mathbf{v} = \mathbf{c}T$. Then

$$\nabla \cdot \mathbf{v} = \mathbf{c} \cdot (\nabla T) + T(\nabla \cdot \mathbf{c}) = \mathbf{c} \cdot (\nabla T)$$

By the divergence theorem and pulling the constant vector out of the integral,

$$\mathbf{c} \cdot \int_V (\nabla T) \, d\tau = \mathbf{c} \cdot \oint_S T \, d\mathbf{a}$$

Since $\mathbf{c}$ can be any arbitrary constant vector, this forces equality of the (vector) integrals.

- (b) Compute

$$\nabla \cdot (\mathbf{v} \times \mathbf{c}) = \mathbf{c} \cdot (\nabla \times \mathbf{v}) - \mathbf{v} \cdot (\nabla \times \mathbf{c}) = \mathbf{c} \cdot (\nabla \times \mathbf{v})$$

By the divergence theorem,

$$\mathbf{c} \cdot \int_V (\nabla \times \mathbf{v}) \, d\tau = \oint_S (\mathbf{v} \times \mathbf{c}) \cdot d\mathbf{a}$$

Rewrite the integrand of the right term as

$$-(\mathbf{c} \times \mathbf{v}) \cdot d\mathbf{a}$$

Since dot products commute, this is equal to

$$-d\mathbf{a} \cdot (\mathbf{c} \times \mathbf{v}) = -\mathbf{c} \cdot (\mathbf{v} \times d\mathbf{a})$$

Therefore our divergence theorem undergoes a makeover to get

$$\mathbf{c} \cdot \int_V (\nabla \times \mathbf{v}) d\tau = -\mathbf{c} \cdot \oint_S \mathbf{v} \times d\mathbf{a}$$

And from the argument of arbitrary $\mathbf{c}$ in the previous derivation, we derive the desired result.

(c) Let $\mathbf{v} = T\nabla U$. Then

$$\nabla \cdot T\nabla U = T\nabla^2 U + (\nabla T) \cdot (\nabla U)$$

By the divergence theorem,

$$\int_V [T\nabla^2 U + (\nabla T) \cdot (\nabla U)] d\tau = \oint_S (T\nabla U) \cdot d\mathbf{a}$$

(d) Subtract

$$\int_V [U\nabla^2 T + (\nabla T) \cdot (\nabla U)] d\tau = \oint_S (U\nabla T) \cdot d\mathbf{a}$$

from the result in (c).

(e) Let $\mathbf{v} = \mathbf{c}T$. Its curl is

$$\nabla \times \mathbf{c}T = T(\nabla \times \mathbf{c}) - \mathbf{c} \times (\nabla T) = -\mathbf{c} \times (\nabla T)$$

which, when dotted with $d\mathbf{a}$, produces

$$d\mathbf{a} \cdot [-\mathbf{c} \times (\nabla T)] = -\mathbf{c} \cdot (\nabla T \times d\mathbf{a})$$

By Stokes' theorem, we get

$$\mathbf{c} \cdot \int_S \nabla T \times d\mathbf{a} = -\mathbf{c} \cdot \oint_{\mathcal{P}} T d\mathbf{l}$$

Once again, by the arbitrariness of $\mathbf{c}$, it must be true that the integrals are equal.

Problem: 1.62

The integral

$$\mathbf{a} \equiv \int_{\mathcal{S}} d\mathbf{a}$$

is sometimes called the **vector area** of the surface $\mathcal{S}$. If $\mathcal{S}$ happens to be *flat*, then $|\mathbf{a}|$ is the *ordinary* (scalar) area, obviously.

- (a) Find the vector area of a hemispherical bowl of radius R .
- (b) Show that $\mathbf{a} = \mathbf{0}$ for any *closed* surface. [*Hint*: Use Prob. 1.61a.]
- (c) Show that $\mathbf{a}$ is the same for all surfaces sharing the same boundary.
- (d) Show that

$$\mathbf{a} = \frac{1}{2} \oint \mathbf{r} \times d\mathbf{l}$$

where the integral is around the boundary line. [*Hint*: One way to do it is to draw the cone subtended by the loop at the origin. Divide the conical surface up into infinitesimal triangular wedges, each with vertex at the origin and opposite side $d\mathbf{l}$, and exploit the geometrical interpretation of the cross product (Fig 1.8).]

- (e) Show that

$$\oint (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l} = \mathbf{a} \times \mathbf{c}$$

for any constant vector $\mathbf{c}$. [*Hint*: Let $T = \mathbf{c} \cdot \mathbf{r}$ in Prob. 1.61e.]

- (a) Begin by recalling the infinitesimal vector area for the surface of a sphere:

$$d\mathbf{a} = r^2 \sin(\theta) d\theta d\phi \hat{\mathbf{r}}$$

This is not simply a matter of integrating, however, as $\hat{\mathbf{r}}$ is itself a function of θ and ϕ . Thus the full integrand is written as

$$d\mathbf{a} = (r^2 \sin(\theta) d\theta d\phi) (\sin(\theta) \cos(\phi) \hat{\mathbf{x}} + \sin(\theta) \sin(\phi) \hat{\mathbf{y}} + \cos(\theta) \hat{\mathbf{z}})$$

One cost-saving observation we can make is imagining around the face of the sphere – **when broken into their $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components, the anti-parallel components cancel**. Only the vertical $\hat{\mathbf{z}}$ components remain. We can verify this is the case by examining how the $\cos(\phi)$ and $\sin(\phi)$ terms integrate from 0 to 2π : the $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ terms vanish.

Simply, we compute (setting $r = R$)

$$\begin{aligned} \mathbf{a} &= \int_0^{2\pi} \int_0^{\pi/2} R^2 \sin(\theta) \cos(\theta) d\theta d\phi \hat{\mathbf{z}} \\ &= \pi R^2 \hat{\mathbf{z}} \end{aligned}$$

- (b) From the expression derived in problem 1.61(a),

$$\mathbf{c} \cdot \int_{\mathcal{V}} (\nabla T) d\tau = \mathbf{c} \cdot \oint_{\mathcal{S}} T d\mathbf{a}$$

for any arbitrary constant $\mathbf{c}$. Let $T = 1$. Then immediately we deduce

$$\mathbf{c} \cdot \int_V \mathbf{0} d\tau = \mathbf{c} \cdot \oint_S d\mathbf{a} = 0$$

But since this must hold for any arbitrary $\mathbf{c}$, it follows that

$$\oint_S d\mathbf{a} = \mathbf{0}$$

Once more, the physical intuition of cancellation holds. When outward normal vectors to the surface are decomposed into their components, all will cancel.

- (c) Consider two surfaces S_1 and S_2 that share the same boundary. **Together, they form a closed surface**, so from part (b), we have

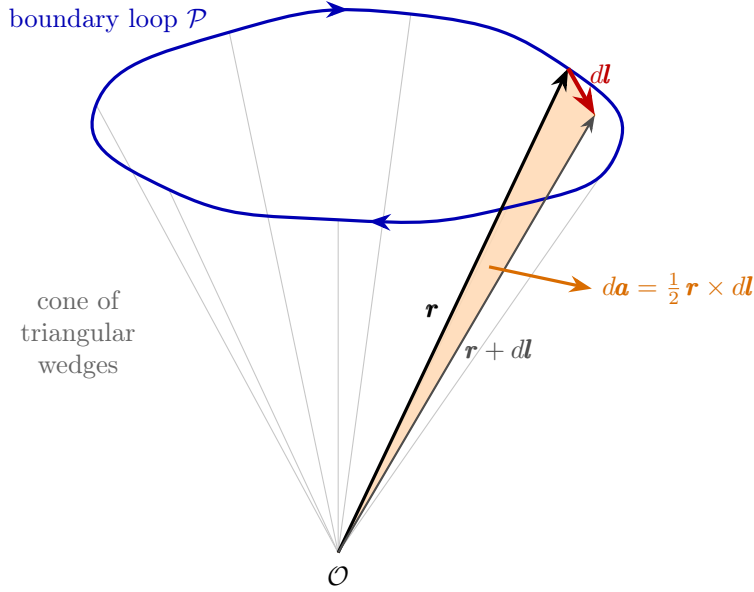
$$\int_{S_1} d\mathbf{a}_1 - \int_{S_2} d\mathbf{a}_2 = \mathbf{0}$$

Some explanation is needed on the negative sign. In fusing the surfaces along a common boundary line and fixing the right-hand rule in one direction around the loop, the direction of $d\mathbf{a}$ is established. However, where $d\mathbf{a}$ will agree with the outward normal on one of the surfaces, it will run opposite to the outward normal of the other. The second surface therefore contributes $-\mathbf{a}_2$ to the closed surface's vector area, which must vanish per part (b). Thus we can conclude

$$\int_{S_1} d\mathbf{a}_1 = \int_{S_2} d\mathbf{a}_2$$

Note: One of the conceptual difficulties of this problem is Griffiths does not offer much in terms of language to capture the subtleties involved here.

- (d) The hint is illustrated below. The idea is to take an arbitrary loop and draw a cone from connecting the origin to the loop at all points via straight lines.



Geometrically, the magnitude of the cross product represents the area of the parallelogram spanned by the two vectors. In this case, we deal with triangular wedges, producing the $1/2$ factor. The triangle, per the diagram, is spanned by $\mathbf{r}$ and $\mathbf{r} + d\mathbf{l}$, from which we calculate

$$d\mathbf{a} = \frac{1}{2} \mathbf{r} \times (\mathbf{r} + d\mathbf{l}) = \frac{1}{2} (\mathbf{r} \times \mathbf{r} + \mathbf{r} \times d\mathbf{l}) = \frac{1}{2} \mathbf{r} \times d\mathbf{l}$$

which represents the infinitesimal vector area of an arbitrary triangular wedges; summing all such $d\mathbf{a}$'s produces

$$\mathbf{a} = \frac{1}{2} \oint \mathbf{r} \times d\mathbf{l}$$

which is true for **all** surfaces sharing the loop per part (c).

(e) The result of problem 1.61(e) is

$$\int_S \nabla T \times d\mathbf{a} = - \oint_P T d\mathbf{l}$$

Let $T = \mathbf{c} \cdot \mathbf{r}$. Then let $\mathbf{c} = (c_1, c_2, c_3)$ and $\mathbf{r} = (x, y, z)$. Their dot product is

$$\mathbf{c} \cdot \mathbf{r} = c_1 x + c_2 y + c_3 z$$

And its gradient produces

$$\nabla(\mathbf{c} \cdot \mathbf{r}) = (c_1, c_2, c_3) = \mathbf{c}$$

Then we can write

$$\int_S \mathbf{c} \times d\mathbf{a} = \mathbf{c} \times \int_S d\mathbf{a} = \mathbf{c} \times \mathbf{a} = - \oint_P (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l}$$

Multiply by a negative sign in the last equality and use the anti-commutativity property of the cross product to conclude

$$\mathbf{a} \times \mathbf{c} = \oint_{\mathcal{P}} (\mathbf{c} \cdot \mathbf{r}) d\mathbf{l}$$

Problem: 1.63

- (a) Find the divergence of the function

$$\mathbf{v} = \frac{\hat{\mathbf{r}}}{r}$$

First compute it directly, as in Eq. 1.84. Test your result using the divergence theorem, as in Eq. 1.85. Is there a delta function at the origin, as there was for $\hat{\mathbf{r}}/r^2$? What is the general formula for the divergence of $r^n \hat{\mathbf{r}}$?

- (b) Find the *curl* of $r^n \hat{\mathbf{r}}$. Test your conclusion using Prob. 1.61b.

- (a) Compute

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \frac{1}{r} \right) = \frac{1}{r^2}$$

Using the divergence theorem over the sphere of radius R , we have

$$\begin{aligned} \int_V \nabla \cdot \mathbf{v} d\tau &= \int_0^{2\pi} \int_0^\pi \int_0^R \left(\frac{1}{r^2} \right) \cdot r^2 \sin(\theta) dr d\theta d\phi \\ &= 4\pi R \\ \oint_S \mathbf{v} \cdot d\mathbf{a} &= \int \left(\frac{1}{R} \hat{\mathbf{r}} \right) \cdot (R^2 \sin(\theta) d\theta d\phi \hat{\mathbf{r}}) \\ &= \int_0^{2\pi} \int_0^\pi R \sin(\theta) d\theta d\phi \\ &= 4\pi R \end{aligned}$$

In this instance, there is no delta function, as both the volume and surface integrals agree in the divergence theorem. In the case of $\mathbf{v} = \hat{\mathbf{r}}/r^2$, the divergence is concentrated at the origin, so the delta function was warranted to account for that missing contribution.

In general,

$$\nabla \cdot r^n \hat{\mathbf{r}} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot r^n) = (n+2) r^{n-1}, \quad n > -2$$

where at $n = -2$, the divergence is $4\pi\delta^3(\mathbf{r})$ as previously derived. For $n < -2$, observe (using the case of $n = -3$):

$$\nabla \cdot \frac{1}{r^3} \hat{\mathbf{r}} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \frac{1}{r^3} \right) = -\frac{1}{r^4}$$

Upon integrating, we find

$$\int_V \left(\nabla \cdot \frac{1}{r^3} \hat{\mathbf{r}} \right) d\tau \rightarrow -\infty$$

So the integral of the divergence diverges, and is ill-defined for the aforementioned values of n .

- (b) Since the $\hat{\mathbf{r}}$ component is purely a function of r (namely, independent of θ and ϕ), and the $\hat{\theta}$ and $\hat{\phi}$ components vanish, $\nabla \times r^n \hat{\mathbf{r}} = \mathbf{0}$. Using the result from problem 1.61(b):

$$\int_V (\nabla \times \mathbf{v}) d\tau = - \oint_S \mathbf{v} \times d\mathbf{a}$$

Since $\hat{\mathbf{r}}$ and $d\mathbf{a}$ are parallel, the integrand in the surface equation also vanishes. The crux of this problem is a parallel of part (a) where the $n = -2$ case misses the divergence contribution at the origin, which we dealt with using the delta function. In this case, we have no such equivalent for the *curl* of $r^n \hat{\mathbf{r}}$, even in the $n = -2$ case.

Problem: 1.64

In case you're not persuaded that $\nabla^2(1/r) = -4\pi\delta^3(\mathbf{r})$ (Eq. 1.102 with $\mathbf{r}' = 0$ for simplicity), try replacing r by $\sqrt{r^2 + \epsilon^2}$, and watching what happens as $\epsilon \rightarrow 0$. Specifically, let

$$D(r, \epsilon) \equiv -\frac{1}{4\pi} \nabla^2 \frac{1}{\sqrt{r^2 + \epsilon^2}}$$

To demonstrate that this goes to $\delta^3(\mathbf{r})$ as $\epsilon \rightarrow 0$:

- (a) Show that $D(r, \epsilon) = (3\epsilon^2/4\pi)(r^2 + \epsilon^2)^{-5/2}$.
- (b) Check that $D(0, \epsilon) \rightarrow \infty$, as $\epsilon \rightarrow 0$.
- (c) Check that $D(r, \epsilon) \rightarrow 0$, as $\epsilon \rightarrow 0$, for all $r \neq 0$.
- (d) Check that the integral of $D(r, \epsilon)$ over all space is 1.

- (a) From the spherical Laplacian, compute

$$\begin{aligned} \nabla^2 \frac{1}{\sqrt{r^2 + \epsilon^2}} &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} (r^2 + \epsilon^2)^{-1/2} \right) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot (-r) (r^2 + \epsilon^2)^{-3/2} \right) \\ &= -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^3 (r^2 + \epsilon^2)^{-3/2} \right) \\ &= -\frac{1}{r^2} \left[3r^2 (r^2 + \epsilon^2)^{-3/2} - 3r^4 (r^2 + \epsilon^2)^{-5/2} \right] \\ &= -3(r^2 + \epsilon^2)^{-3/2} + 3r^2 (r^2 + \epsilon^2)^{-5/2} \\ &= -3(r^2 + \epsilon^2)(r^2 + \epsilon^2)^{-5/2} + 3r^2 (r^2 + \epsilon^2)^{-5/2} \\ &= -3\epsilon^2 (r^2 + \epsilon^2)^{-5/2} \end{aligned}$$

Substituting into $D(r, \epsilon)$ produces the desired result:

$$D(r, \epsilon) = \frac{3\epsilon^2}{4\pi} (r^2 + \epsilon^2)^{-5/2}$$

- (b) We have

$$D(0, \epsilon) = \frac{3\epsilon^2}{4\pi} \frac{1}{\epsilon^5} = \frac{3}{4\pi\epsilon^3}$$

From which it is immediately apparent

$$\lim_{\epsilon \rightarrow 0} D(0, \epsilon) = \infty$$

- (c) For non-zero r we have $\epsilon \ll r$, so approximate

$$D(r, \epsilon) \approx \frac{3\epsilon^2}{4\pi r^5}$$

which vanishes as $\epsilon \rightarrow 0$.

- (d) This is not the easiest integral to compute, but it is tractable. Before us we have

$$\int_{\mathbb{R}^3} D(r, \epsilon) d\tau = \int_0^{2\pi} \int_0^\pi \int_0^\infty \frac{3\epsilon^2}{4\pi} \frac{r^2 \sin(\theta)}{(r^2 + \epsilon^2)^{5/2}} dr d\theta d\phi$$

Solving over the θ and ϕ bounds, simplify to

$$\int_0^\infty \frac{3r^2 \epsilon^2}{(r^2 + \epsilon^2)^{5/2}} dr$$

Now, we will need to make a u -substitution. Let $u = r/\epsilon$. Then $du = dr/\epsilon$, and we rewrite

$$\int_0^\infty \frac{3u^2 \epsilon^5}{(u^2 \epsilon^2 + \epsilon^2)^{5/2}} du = 3 \int_0^\infty \frac{u^2}{(1 + u^2)^{5/2}} du$$

Next, we make yet another substitution, this time of the trigonometric variety. Let $u = \tan(\psi)$, then $du = \sec^2(\psi) d\psi$. Note that the bounds of integration change to $[0, \pi/2)$. We further simplify to

$$3 \int_0^{\pi/2} \frac{\tan^2(\psi)}{\sec^5(\psi)} \sec^2(\psi) d\psi = 3 \int_0^{\pi/2} \frac{\tan^2(\psi)}{\sec^3(\psi)} d\psi = 3 \int_0^{\pi/2} \sin^2(\psi) \cos(\psi) d\psi$$

And finally, we can compute

$$3 \int_0^{\pi/2} \sin^2(\psi) \cos(\psi) d\psi = \sin^3(\psi) \Big|_0^{\pi/2} = 1$$

Since $\delta^3(\mathbf{r})$ is best thought of as a **generalized function** and not as a function per se, and has the property that it must integrate to 1 over all space, we can now articulate the argument that $D(r, \epsilon) \rightarrow \delta^3(\mathbf{r})$: from part (b), $D(r, \epsilon)$ spikes at the origin; from part (c), $D(r, \epsilon)$ goes to zero everywhere else; and from this part, $D(r, \epsilon)$ has an integral of 1 over all space.